



SEEK WISDOM, ELEVATE YOUR INTELLECT AND SERVE HUMANITY !

Addis Ababa University
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Addis Ababa Institute of Technology

School of Mechanical and Industrial Engineering

Engineering Materials I

(MEng 2122)

Chapter - Two

Atomic Structure and Interatomic Bonding

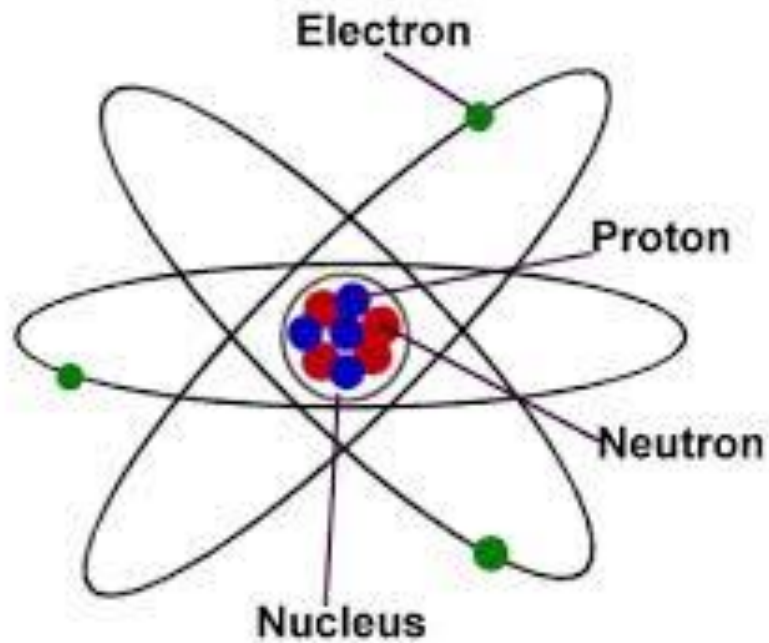
Atomic Structure and Interatomic Bonding

Content - Outline

- Introduction
- Atomic Structure
 - ✓ Basic Concepts of Atomic Structure
 - ✓ Periodic Table
 - ✓ Electron Configuration in Atoms
- Atomic Bonding in Solids
 - ✓ Bonding Forces and Energies
 - ✓ Primary Interatomic Bonds
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Atomic Structure & Interatomic Bonding

Introduction to Chemistry of Materials



*A gecko can support its body mass with a single toe! This remarkable ability is due to the presence of an extremely large number of microscopically small hairs on each of their toe pads. By **van der Waals forces** between hair molecules and surface.*

Learning Objectives

After completing this chapter, you should be able to:

1. State the two atomic models cited, and note the differences between them.
2. Describe the important quantum-mechanical principle that relates to electron energies.
3. Schematically plot attractive, repulsive, and net energies versus interatomic separation for two atoms or ions.
4. Understand the equilibrium separation and the bonding energy.
5. Briefly describe ionic, covalent, metallic, hydrogen, and van der Waals bonds.
6. Identify which materials exhibit each of these bonding types.

Atomic Structure & Interatomic Bonding

*Why we Study Atomic Structure
and Interatomic Bonding?*



Introduction

Why we Study Atomic Structure and Interatomic Bonding?

- An important reason to have an understanding of **interatomic bonding** in solid is that in some instances, the type of bond allows us to explain a **material's properties**.
- Some of the important properties of **solid materials** depend on;
 - geometric **atomic** arrangements, and
 - **interactions** that exist among constituent atoms or molecules.
- Several **fundamental** and **important concepts** will be discussed in this chapter like;
 - atomic structure, electron configurations in atoms,
 - periodic table, and
 - various types of primary and secondary interatomic bonds.

Atomic Structure

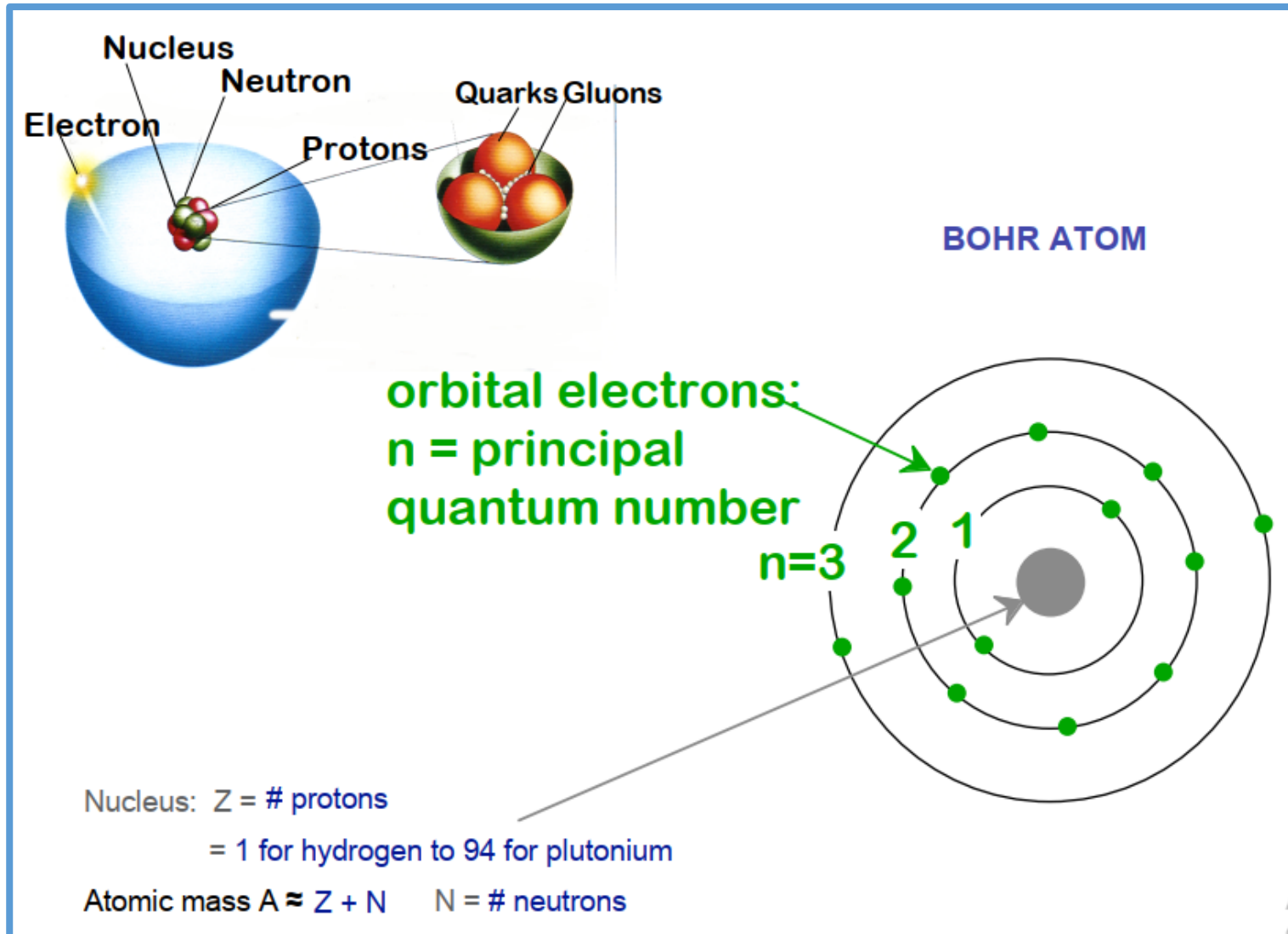
Basic Concepts

- All **matters** are composed of **atoms**.
- Understanding the **structure of atoms** is critical to understand the **nature** of **biological molecules**.
- An **atom** consists of;
 - a very small **nucleus**: composed of **protons** and **neutrons**
 - And an **electrons** encircling the nucleus.
- Both **electrons** and **protons** are electrically charged.
 - Their magnitude is; $1.602 \times 10^{-19} \text{C}$,
 - **negative** in sign for **electrons** and
 - **positive** for **protons**
 - **neutrons** are electrically **neutral**.
- **Masses** for **protons** and **neutrons** are extremely small; have approximately the same mass, $1.67 \times 10^{-27} \text{kg}$,

Atomic Structure

Basic Concepts

- Mass of an electron is; 9.11×10^{-31} kg
- Protons and neutrons are located in the nucleus.
- Electrons are found in orbitals surrounding the nucleus.
- Each chemical element is characterized by the number of protons in the nucleus, or the *atomic number* (Z).
- For a neutral or complete atom, the *atomic number* equals to the number



Atomic Structure

Basic Concepts

- The number of **proton** is same with the atomic number (Z).
- The number of **neutrons** (N) may be variable. Thus atoms of some elements have two or more **different atomic masses**, which are called **isotopes**.
- The **atomic weight** of an element corresponds to the **weighted average** of the atomic masses of the atom's naturally occurring **isotopes**. $A = Z + N$
- The **atomic mass unit (amu)** may be used to compute atomic weight where
1 amu is defined as 1/12 of the atomic mass of the most common isotope of carbon, carbon 12 ($^{12}_C$) ($A=12.0$). Within this scheme, the masses of protons and neutrons are slightly greater than unity
 $1 \text{ amu/atom (or molecule)} = 1 \text{ g/mol}$
For example, the atomic weight of iron is 55.85 amu/atom, or 55.85g/mol.

Atomic Structure

Basic Concepts

- One **mole** of a substance is,
 6.022×10^{23} (Avogadro's number) atoms or molecules.
- Atoms with the **same atomic number** have the same chemical properties and belong to the same **element**.
- **Isotopes** – atoms of the same element that have different atomic mass numbers due to different numbers of neutrons.

For example: see isotopes of carbon

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Carbon-12	Carbon-13	Carbon-14
6 Protons 6 Neutrons 6 Electrons	6 Protons 7 Neutrons 6 Electrons	6 Protons 8 Neutrons 6 Electrons
		

Atomic Structure

EXAMPLE PROBLEM 2.1

Average Atomic Weight Computation for Cerium

Cerium has four naturally occurring isotopes: 0.185% of ^{136}Ce , with an atomic weight of 135.907 amu; 0.251% of ^{138}Ce , with an atomic weight of 137.906 amu; 88.450% of ^{140}Ce , with an atomic weight of 139.905 amu; and 11.114% of ^{142}Ce , with an atomic weight of 141.909 amu. Calculate the average atomic weight of Ce.

Atomic Structure

Solution

The average atomic weight of a hypothetical element M, $\bar{A}_M$, is computed by adding fraction-of-occurrence — atomic weight products for all its isotopes; that is,

$$\bar{A}_M = \sum_i f_{iM} A_{iM} \quad (2.2)$$

In this expression, f_{iM} is the fraction-of-occurrence of isotope i for element M (i.e., the percentage-of-occurrence divided by 100), and A_{iM} is the atomic weight of the isotope.

For cerium, Equation 2.2 takes the form

$$\bar{A}_{\text{Ce}} = f_{136\text{Ce}} A_{136\text{Ce}} + f_{138\text{Ce}} A_{138\text{Ce}} + f_{140\text{Ce}} A_{140\text{Ce}} + f_{142\text{Ce}} A_{142\text{Ce}}$$

Incorporating values provided in the problem statement for the several parameters leads to

$$\begin{aligned} \bar{A}_{\text{Ce}} &= \left(\frac{0.185\%}{100} \right) (135.907 \text{ amu}) + \left(\frac{0.251\%}{100} \right) (137.906 \text{ amu}) + \left(\frac{88.450\%}{100} \right) (139.905 \text{ amu}) \\ &\quad + \left(\frac{11.114\%}{100} \right) (141.909 \text{ amu}) \\ &= (0.00185)(135.907 \text{ amu}) + (0.00251)(137.906 \text{ amu}) + (0.8845)(139.905 \text{ amu}) \\ &\quad + (0.11114)(141.909 \text{ amu}) \\ &= 140.115 \text{ amu} \end{aligned}$$

Atomic Structure

Electrons in Atoms

Atomic Models:

- Quantum mechanics is a set of principles and laws that govern systems of atomic and subatomic entities.
- An understanding of the behavior of electrons in atoms and crystalline solids necessarily involves the discussion of quantum-mechanical concepts.
- Bohr atomic model, is one early study of quantum mechanics in which electrons are assumed to revolve around the atomic nucleus in discrete orbitals.

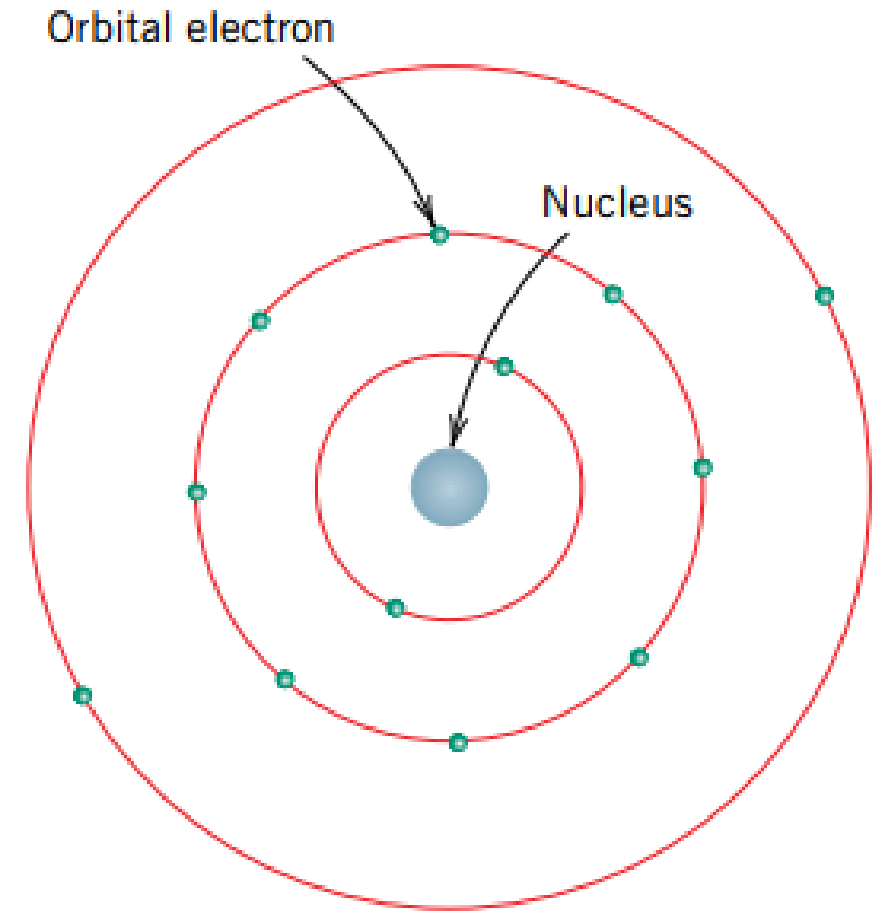


Figure 2.2: Schematic representation of the Bohr atom.

Atomic Structure

Electrons in Atoms

Atomic Models:

- Another important **quantum-mechanical** principle specifies that the energies of electrons are *quantized*:
 - Electrons are permitted to have only **specific** values of energy.
 - An electron may change energy, but in doing so it must make a **quantum jump** either to an allowed higher energy (with absorption of energy) or to a lower energy (with emission of energy).
 - These electron energies are associated with *energy levels* or *states*.
- This Bohr model was eventually found to have some significant **limitations** because of its inability to explain several phenomena involving electrons.
- A resolution was reached with a **wave-mechanical model**, in which the electron is considered to exhibit both *wave-like* and *particle-like* characteristics.
- With this model, an **electron** is no longer treated as a particle moving in a discrete orbital; rather, **position** is considered.
- It is described by the **probability** distribution or electron cloud.

Atomic Structure

Electrons in Atoms

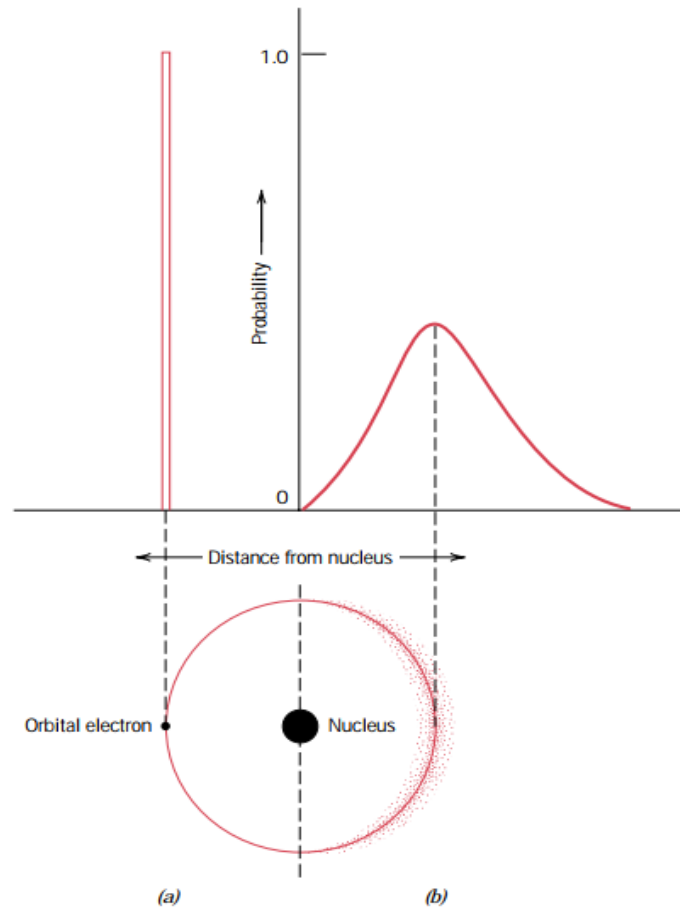


Figure 2.4 Comparison of the (a) Bohr and (b) wave mechanical atom models in terms of electron distribution

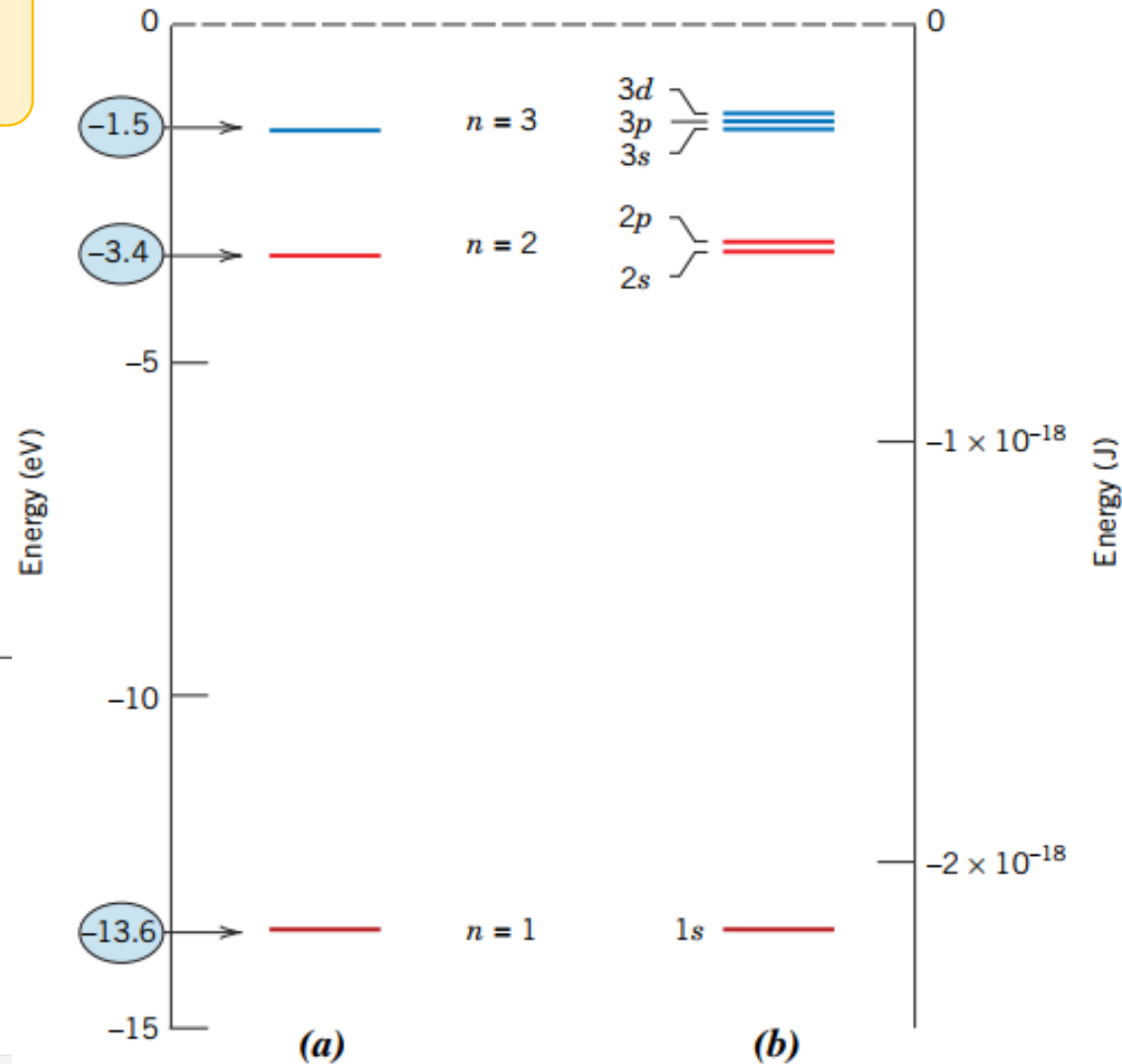


Figure 2.3

(a) The first three electron energy states for the **Bohr hydrogen atom**.

(b) Electron energy states for the first three shells of the **wave mechanical hydrogen atom**.

- The joule equivalent of the electron volt is $1.602 \times 10^{-19} \text{ J} = 1 \text{ eV}$.

Atomic Structure

Electrons in Atoms

Quantum Numbers:

- In **wave mechanics**, **every electron** in an atom is characterized by four parameters called **quantum numbers**.
- The **size**, **shape**, and **spatial** orientation of an electron's probability density are specified by three of these quantum numbers.
- **Bohr energy levels** separate into electron **subshells**, and quantum numbers dictate the number of states within each subshell.

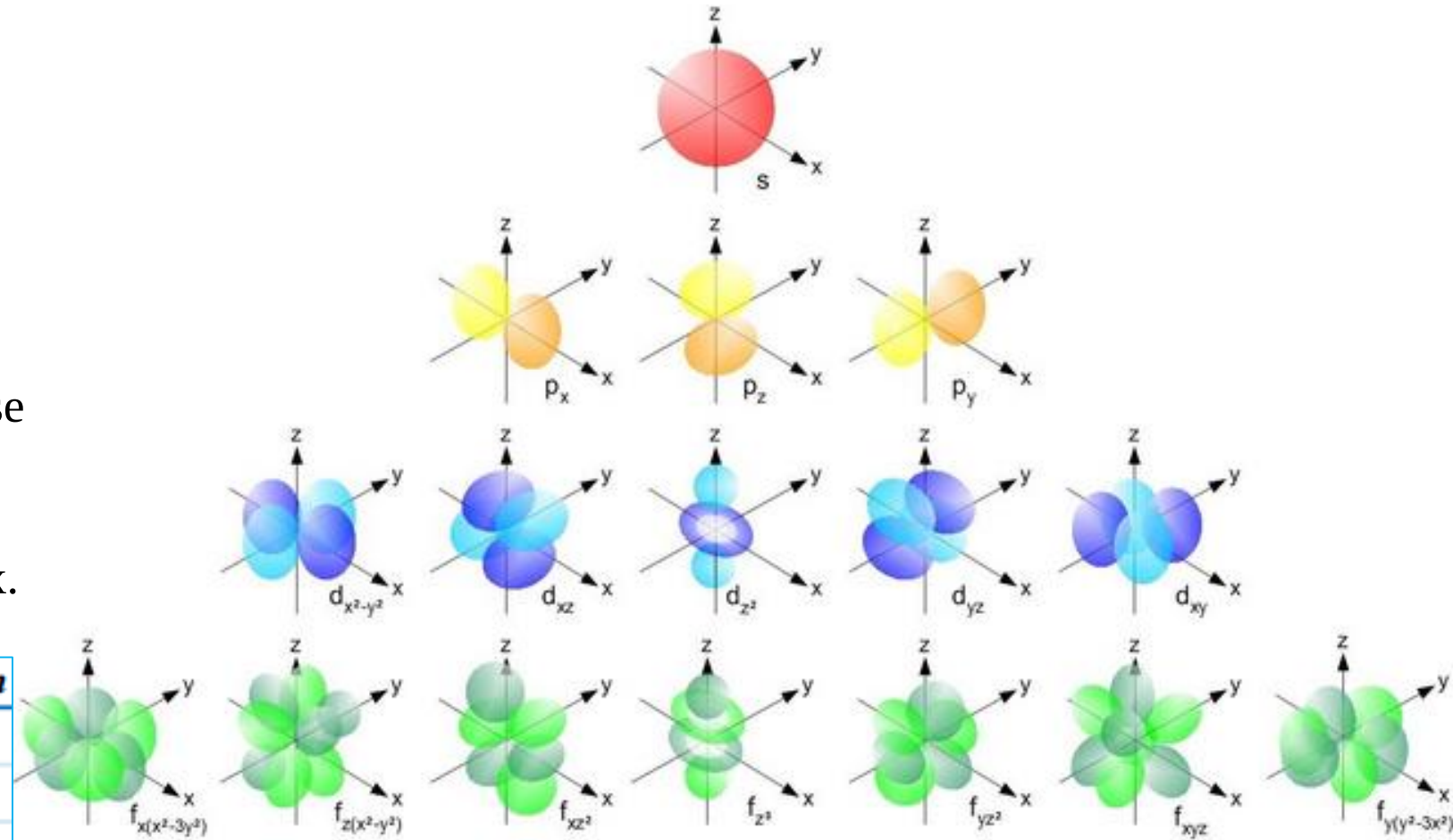
1. **Principal quantum number** n . describes the energy of an electron in the shell.

- These **shells** are designated by the letters K, L, M, N, O , and so on, which correspond, respectively, to $n = 1, 2, 3, 4, 5, \dots$, as indicated in Table 2.1.
2. **Angular or azimuthal** quantum number, l , designates the subshell. Describes the shape of subshell.
- Values of l are restricted by the magnitude of n and can take on integer values that range from $l = 0$ to $l = (n - 1)$.
 - Each **subshell** is denoted by a lowercase letter an s, p, d , or f related to l values as follows:

Atomic Structure

Electron orbital shapes depend on l ;

- s -orbitals are spherical and centered on the nucleus.
- p -subshell orbitals contains three orbitals, these are mutually perpendicular to one another like those of an x - y - z coordinate system.
- d -subshells orbitals are more complex.



Value of l	Letter Designation
0	s
1	p
2	d
3	f

Table 2.1

Atomic Structure

Electrons in Atoms

Table 2.1 Summary of the Relationships among the Quantum Numbers n , l , m_l and Numbers of Orbitals and Electrons

<i>Value of n</i>	<i>Value of l</i>	<i>Values of m_l</i>	<i>Subshell</i>	<i>Number of Orbitals</i>	<i>Number of Electrons</i>
1	0	0	1s	1	2
2	0	0	2s	1	2
	1	-1, 0, +1	2p	3	6
3	0	0	3s	1	2
	1	-1, 0, +1	3p	3	6
	2	-2, -1, 0, +1, +2	3d	5	10
	0	0	4s	1	2
4	1	-1, 0, +1	4p	3	6
	2	-2, -1, 0, +1, +2	4d	5	10
	3	-3, -2, -1, 0, +1, +2, +3	4f	7	14

Atomic Structure

Table 2.1 The Number of Available Electron States in Some of the Electron Shells and Subshells

<i>Principal Quantum Number n</i>	<i>Shell Designation</i>	<i>Subshells</i>	<i>Number of States</i>	<i>Number of Electrons</i>	
				<i>Per Subshell</i>	<i>Per Shell</i>
1	K	s	1	2	2
2	L	s	1	2	8
		p	3	6	
3	M	s	1	2	18
		p	3	6	
		d	5	10	
		s	1	2	
4	N	p	3	6	32
		d	5	10	
		f	7	14	

Atomic Structure

3. **Magnetic quantum number** (m_l); describes the **orientation** or place of orbitals. The **number of electron orbitals** for each subshell is determined by m_l can take on integer values between **$-l$** and **$+l$** , including 0.

- Example, for $l = 1$, $m_l = -1, 0$, and $+1$, and three p orbitals are possible.
- **d -subshells** have five orbitals, and **f -subshells** have seven orbitals.

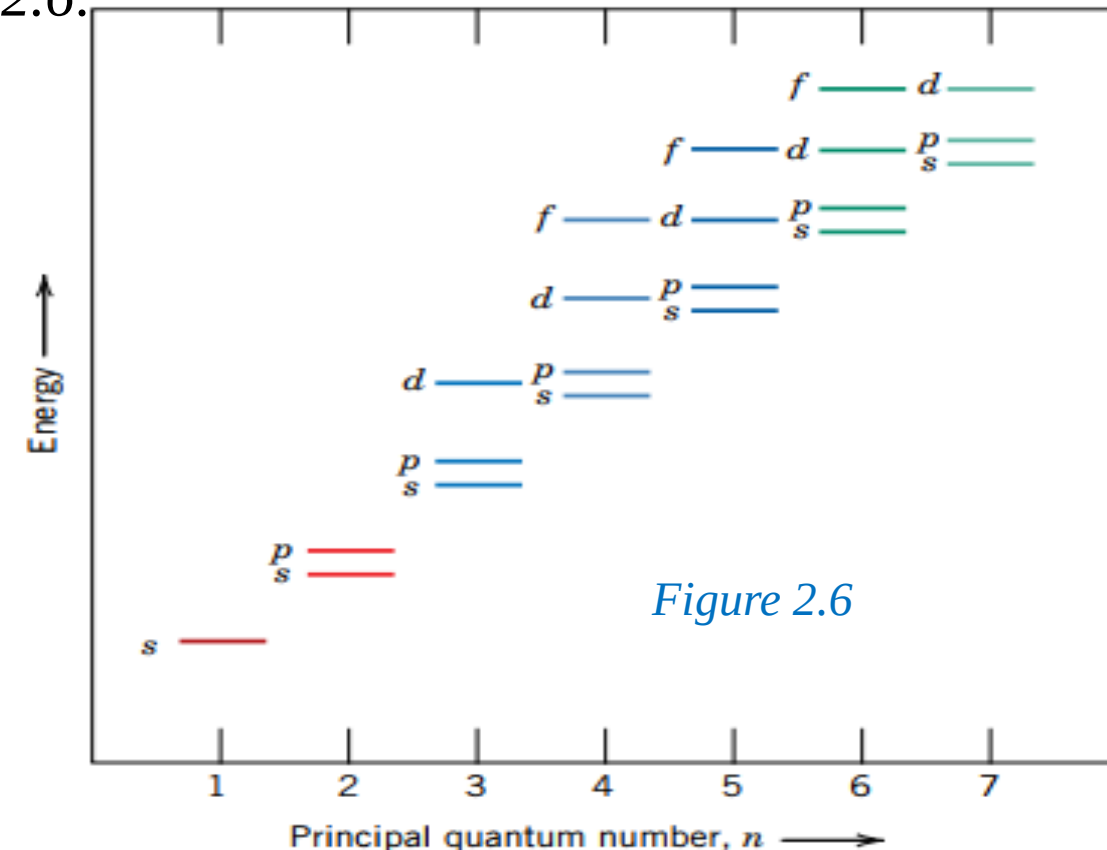
Atomic Structure

Electrons in Atoms

Quantum Numbers:

- In the absence of an **external magnetic field**, all orbitals within each subshell are **identical in energy**.
 - When a **magnetic field** is applied, these subshell states split, with each orbital assuming a slightly **different energy**.
 - A **spin moment** is associated with each electron, which must be oriented either up or down.
4. **Spin moment** is the **fourth quantum number**, m_s , for which two values are possible: $+1/2$ (for spin up) and $-1/2$ (for spin down).

- A **complete energy level diagram** for the various shells and subshells using the **wave-mechanical model** is shown in *Figure 2.6*.



Atomic Structure

- If an atom contains paired electrons in orbitals, one moves anti-clockwise and the other clockwise; the forces produced cancel each other out, i.e. no magnetic property is shown (**Diamagnetic** material).
- If an atom has unpaired electrons, the forces are not canceled and it will create magnetic properties (**Paramagnetic** material).
- The key difference between **paramagnetic** and **diamagnetic** materials is that **paramagnetic** materials are attracted to external magnetic fields, while **diamagnetic** materials repel them. Materials tend to show weak magnetic properties in the presence of an external magnetic field, whereas **ferromagnetic** materials are strongly attracted to an external magnetic field.

Atomic Structure

Exercise

Write the four quantum numbers for an electron in a 3p orbitals?

Tips: 3p_x means

3 -principle quantum number

P -angular quantum number

X -magnetic quantum number

- The Pauli exclusion principle states that in a single atom no two electrons will have an identical set of the same quantum numbers (n , l , m_l , and m_s). In other words, two or more identical fermions cannot occupy the same quantum state within a quantum system simultaneously.*
- A fermion is any particle that has an odd half-integer (like $+1/2$, $-1/2$)*

Atomic Structure

Electrons in Atoms

Electron Configuration:

- *Electron states*: values of energy that are permitted for electrons.
- The *electron configuration* or structure of an atom represents the manner in which these *states* are occupied.
- We use the *Pauli exclusion principle*.
- This is another *quantum-mechanical concept*, which stipulates that each electron state can hold *no more than two electrons* that must have opposite spins.
- Thus, *s*, *p*, *d*, and *f* subshells may each accommodate, a total of **2**, **6**, **10**, and **14** electrons respectively.
- When all the *electrons* occupy the lowest possible energies then an *atom* is said to be in its *ground state*.
- The *valence electrons* are those that occupy the outermost shell.
- The *valence electrons* determines the *group* of an element as well as the *ions*.

Atomic Structure

Electrons in Atoms

Electron Configuration:

Example; electron configuration of H, He, F, Ne, and Na

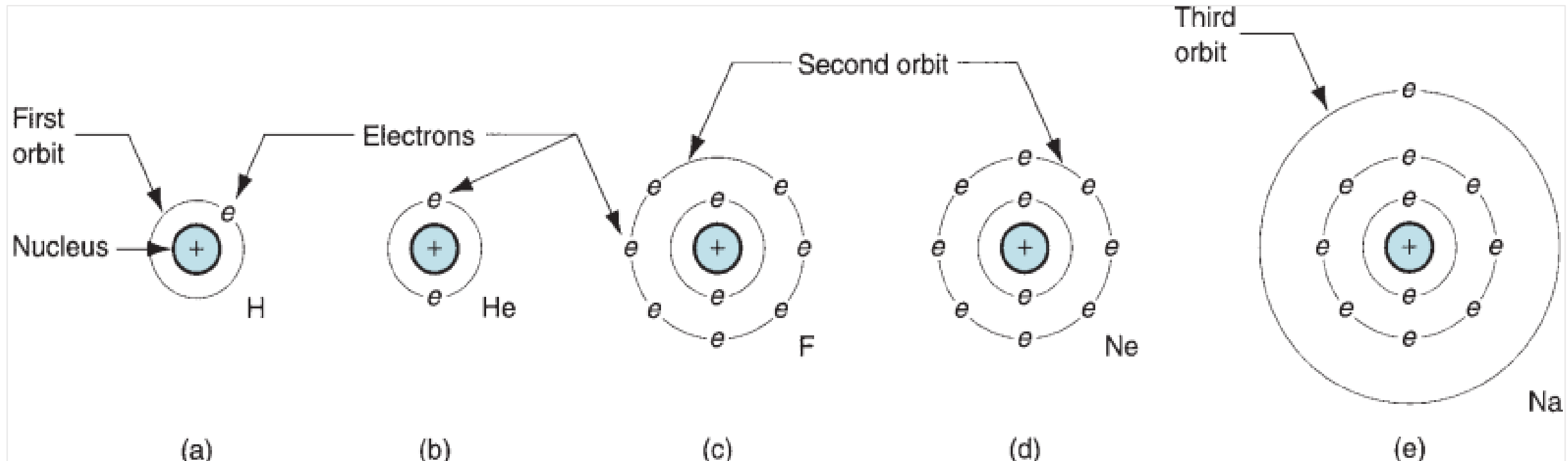


Figure: Simple model of atomic structure for several elements: (a) hydrogen, (b) helium, (c) fluorine, (d) neon, and (e) sodium.

Atomic Structure

Electrons in Atoms

Electron Configuration:

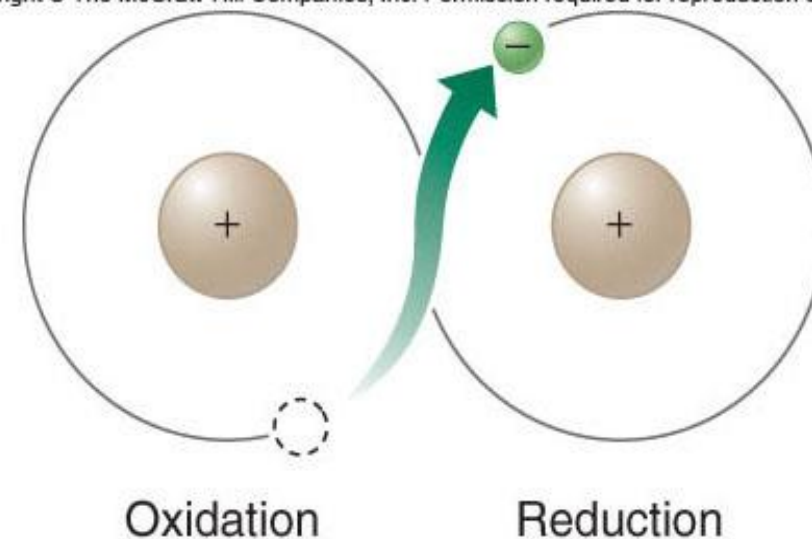
- **Valence electrons** – those in unfilled shells.
- Filled shells are more **stable**.
- **Valence electrons** are most available for bonding and tend to control the the physical and **chemical properties**.

Example: C (atomic number = 6)



- **Electrons** can be transferred from one atom to another, while still retaining the energy of their position in the atom.
 - **oxidation** = loss of an electron
 - **reduction** = gain of an electron

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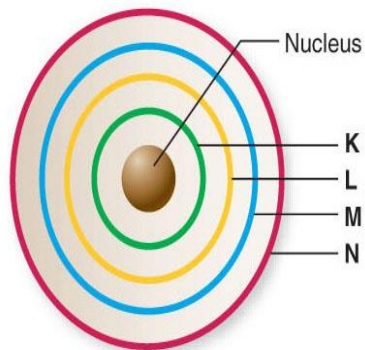
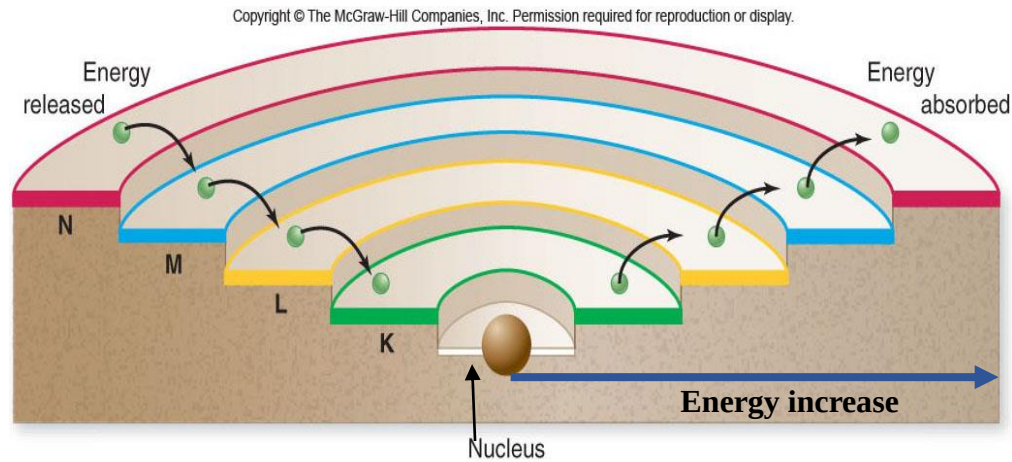
- The **maximum number of electrons** in an orbit = $2n^2$
where n = identifies the orbit, with $n = 1$ closest to the nucleus.

Atomic Structure

Electrons in Atoms

Electron Configuration:

Table 2.2
Expected Electron
Configurations for
Some Common
Elements^a



<i>Element</i>	<i>Symbol</i>	<i>Atomic Number</i>	<i>Electron Configuration</i>
Hydrogen	H	1	$1s^1$
Helium	He	2	$1s^2$
Lithium	Li	3	$1s^2 2s^1$
Beryllium	Be	4	$1s^2 2s^2$
Boron	B	5	$1s^2 2s^2 2p^1$
Carbon	C	6	$1s^2 2s^2 2p^2$
Nitrogen	N	7	$1s^2 2s^2 2p^3$
Oxygen	O	8	$1s^2 2s^2 2p^4$
Fluorine	F	9	$1s^2 2s^2 2p^5$
Neon	Ne	10	$1s^2 2s^2 2p^6$
Sodium	Na	11	$1s^2 2s^2 2p^6 3s^1$
Magnesium	Mg	12	$1s^2 2s^2 2p^6 3s^2$
Aluminum	Al	13	$1s^2 2s^2 2p^6 3s^2 3p^1$
Silicon	Si	14	$1s^2 2s^2 2p^6 3s^2 3p^2$
Phosphorus	P	15	$1s^2 2s^2 2p^6 3s^2 3p^3$
Sulfur	S	16	$1s^2 2s^2 2p^6 3s^2 3p^4$
Chlorine	Cl	17	$1s^2 2s^2 2p^6 3s^2 3p^5$
Argon	Ar	18	$1s^2 2s^2 2p^6 3s^2 3p^6$
Potassium	K	19	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$
Calcium	Ca	20	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$
Scandium	Sc	21	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^1 4s^2$

Atomic Structure

The Periodic Table

- The **periodic table** arranges all elements according to their **atomic number**.
- The table identifies elements with similar **chemical properties**.
- The elements are situated, with increasing **atomic number**.
- The **seven horizontal rows** called **periods**.
- The arrangement is such that all elements arrayed in a given **column** or **group** have similar **valence electron** structures, as well as **chemical** and **physical** properties.
- These properties change gradually, moving **horizontally** across each period and **vertically** down each column.
- The elements positioned in **Group 0** (VIII A), the rightmost group, are the **inert gases**, which have filled electron shells and stable electron configurations. *E.g., He, Ne, Ar, Kr, Xe (Xenon) & Rn (Radon).*
- **Group VIIA** and **VIA** elements are one and two electrons deficient, respectively.
- The **Group VIIA** elements like F, Cl, Br, I, and At are also called **halogens**.
- The **alkali metals** are Li, Na, K, ..., labeled as **Groups IA**. Has one excess electron.
- The **alkaline earth metals** are Be, Mg, Ca, ..., etc., labeled as **Groups IIA**. Has two excess electrons.

Atomic Structure

The Periodic Table:

Periodic Table:

The elements in the three long periods, Groups IIIB through IIB, are termed the *transition metals*.

Figure 2.8 The periodic table of the elements. The numbers in parentheses are the atomic weights of the most stable or common isotopes.

Atomic Structure

The Periodic Table

- *Electronegativity* - how willing atoms are to **accept** electrons.
- Subshells with one electron - *low electronegativity*
- Subshells with one missing electron - *high electronegativity*.

Electronegativity increases from left to right.

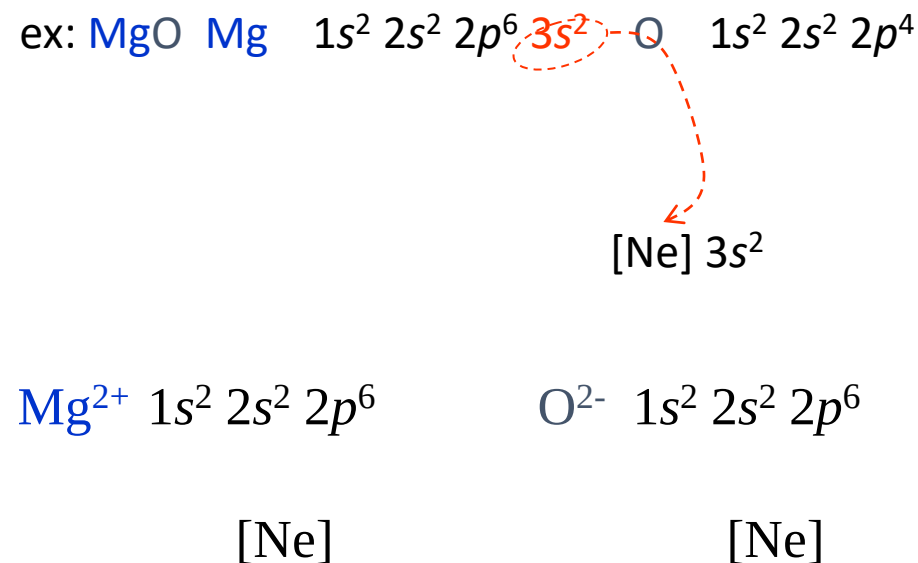
Metals are electropositive – can give up their few valence electrons to become positively charged ions.

Ionic bond – metal + nonmetal

↑ ↑

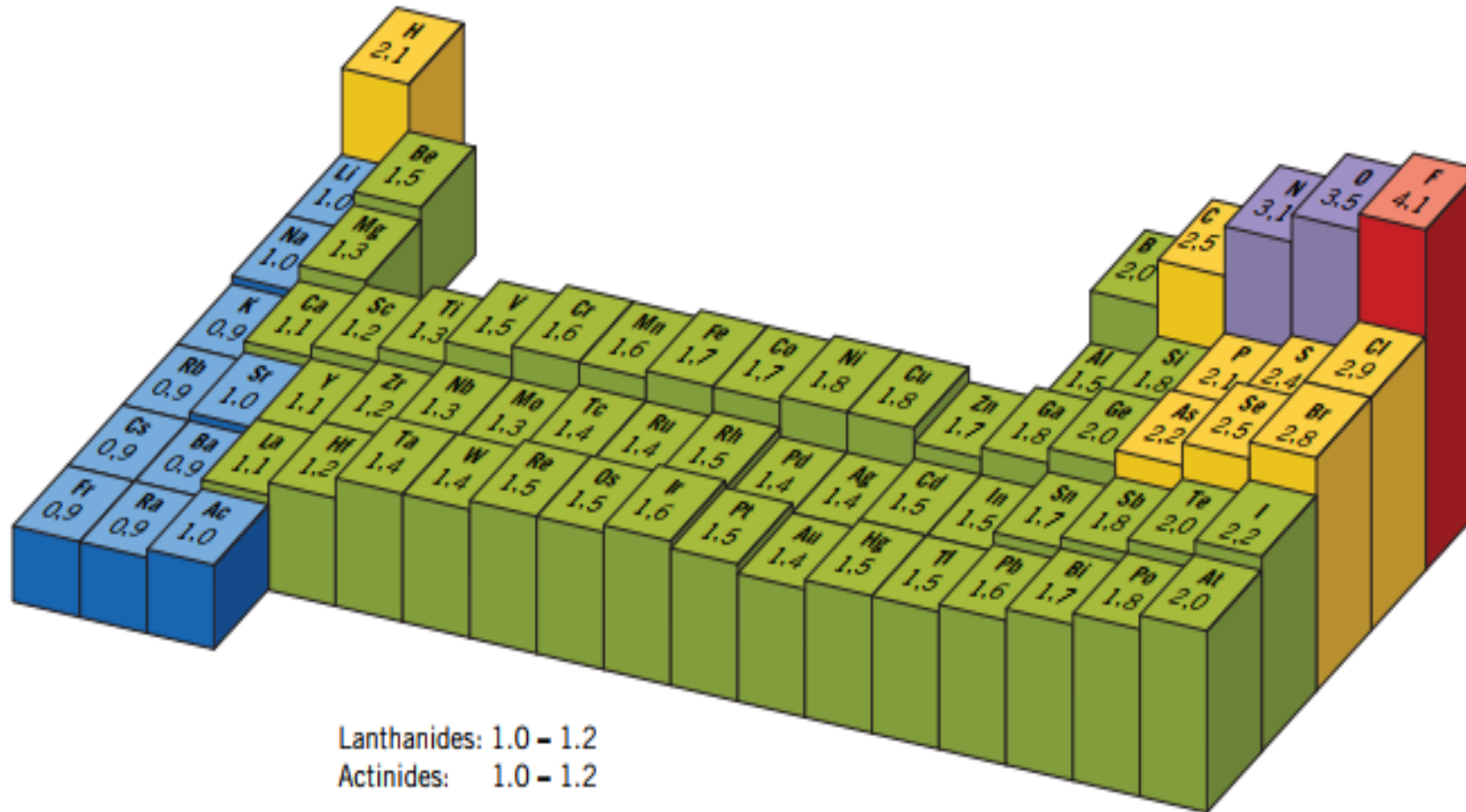
Donates Accept
electrons electrons

Have huge difference in electronegativities.



Atomic Structure

Electronegativity:



- *Electronegativity* is a chemical property that describes the *tendency* of an atom to *attract* a shared pair of electrons towards itself.
- An atom's *electronegativity* is affected by both its *atomic number* and the *distance* at which its valence electrons reside from the charged *nucleus*.

Figure 2.9 The electronegativity values for the elements.

(From J. E. Brady and F. Senese, *Chemistry: Matter and Its Changes*, 4th edition. This material is reproduced with permission of John Wiley & Sons, Inc.)

Atomic Bonding in Solids

Bonding Forces & Energies

- The *physical properties* of materials is predicted on a knowledge of the *interatomic forces* that bind the atoms together.
- At *large distances*, the interactions are *negligible*; but as the atoms approach, each exerts forces on the other.
- These forces are of two types, attractive (F_A) and repulsive (F_R), and the magnitude of each depends on the separation or *interatomic distance* (r).
- The origin of F_A depends on the type of *bonding* exists between the *two atoms*.
- Ultimately, the outer electron shells of the two atoms begin to overlap, and a strong repulsive force F_R comes into play.
- The net force F_N between the two atoms is just;
$$F_N = F_A + F_R$$
 - Repulsion when they are brought close together.
 - (As electron clouds overlap energy increases)
 - Attractive part: at large distances (depends on type of bonding).

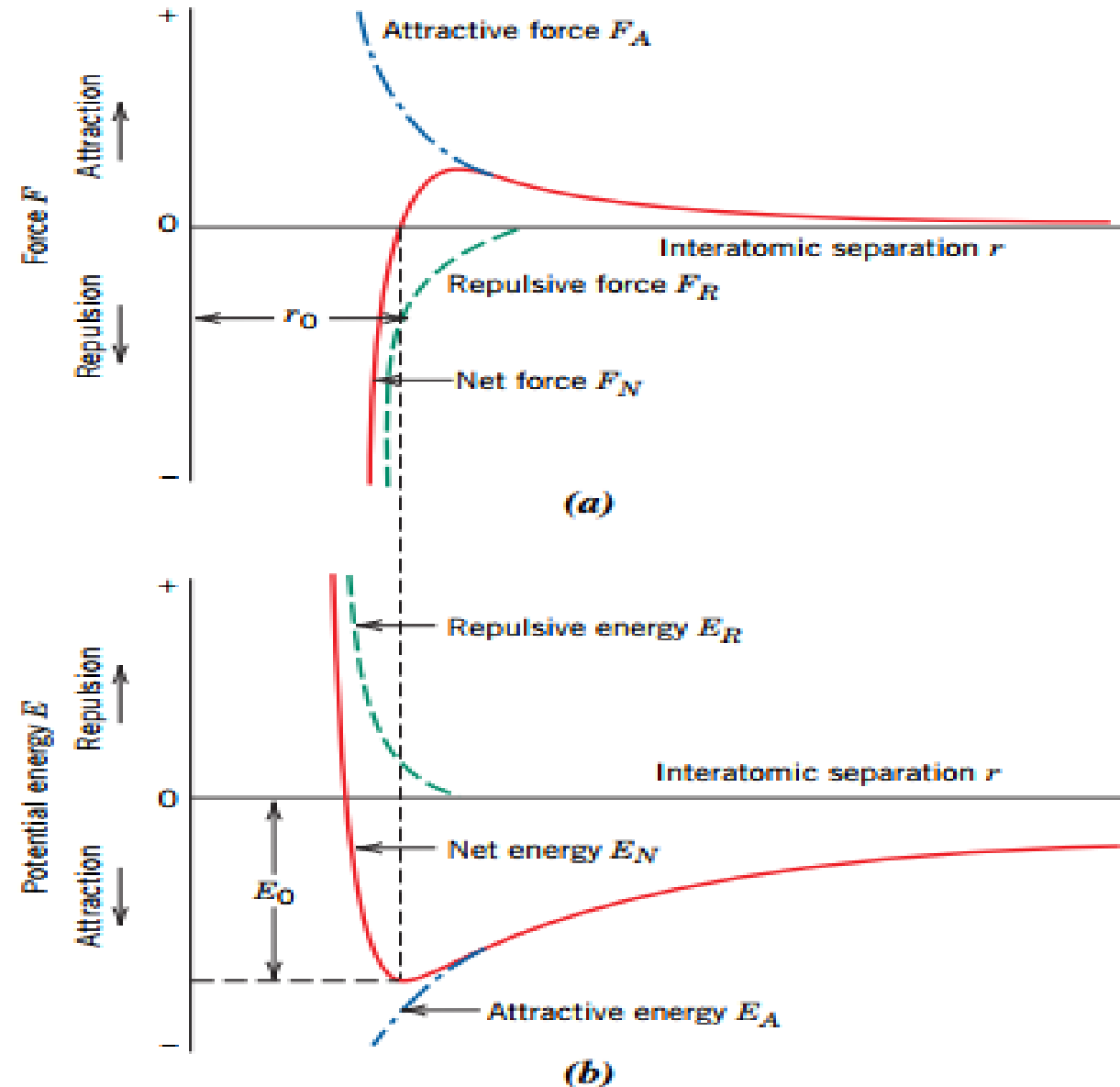
Atomic Bonding in Solids

Bonding Forces & Energies

- Bonding holds atoms together to form solids materials • In solids, atoms are held at preferred distances from each other (equilibrium distances)
- Distances **larger or smaller** than equilibrium distances are not preferred. Equilibrium **spacing** r_0 is approximately 0.3nm
- Consequently, as atomic bonds are **stretched**, atoms tend to **attract** each other, and as the bonds are **compressed**, atoms repel each other.
- Simple bonding models assume that the total bonding results from the sum of two forces: an attractive force (F_A) and a repulsive (F_R).
- At **equilibrium** they are just equal.

When F_A and F_R are equal in magnitude but opposite in sign,

$$F_A + F_R = 0$$



Atomic Bonding in Solids

Bonding Forces & Energies

- Sometimes it is more convenient to work with the **potential energies** between two atoms instead of forces. Mathematically, Energy (E) and force (F) are related as;

And, for atomic systems,

$$E = \int F dr$$

$$\begin{aligned} E_N &= \int_r^\infty F_N dr \\ &= \int_r^\infty F_A dr + \int_r^\infty F_R dr \\ &= E_A + E_R \end{aligned}$$

- The **minimum** in the net energy curve corresponds to the equilibrium spacing, r_0 .

- The **bonding energy** for these two atoms, E_0 , corresponds to the energy at this minimum point.

Electron volt – energy lost / gained when an electron is taken through a potential difference of one volt.

$$E = q \times V$$

For $q = 1.6 \times 10^{-19}$ Coulombs
 $V = 1$ volt

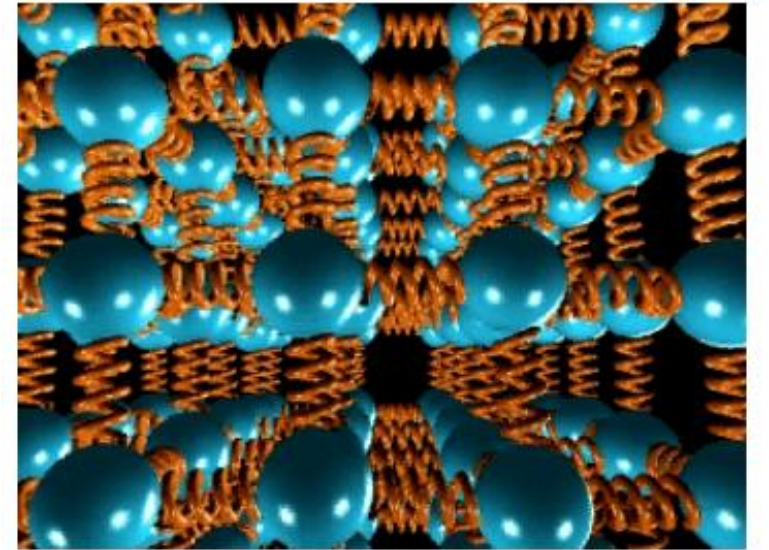
$$1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$$

Atomic Bonding in Solids

Chemical Bonding

Interatomic bonds:

- forces that hold atoms together which act like little springs, linking one atom to the next in the solid state.



- Interatomic Forces:

- attractive forces (F_a)
- repulsive forces (F_r)

The Coulomb's law:

$$F = K_e \cdot \frac{q_1 \cdot q_2}{r^2}$$

- ✓ q_1 : Charge of object 1
- ✓ q_2 : Charge of object 2
- ✓ r : Distance between the two objects
- ✓ $K_e = 8.9875 \times 10^9 \text{ N.m}^2.\text{C}^{-2}$ (Coulomb Constant)
- ✓ F : Force between the two objects. $F > 0$ implies a repulsive interaction, while $F < 0$ means an attractive interaction.

Atomic Bonding in Solids

Chemical Bonding

Interatomic bonds:

- Force-Energy Relationship:

$$F = -\frac{dV}{dr}$$

“Force is the negative gradient of potential energy”

Energy calculation:

$$V_N = \int_{\infty}^r F_A(r)dr + \int_{\infty}^r F_R(r)dr$$

Equilibrium is reached by minimizing V_N

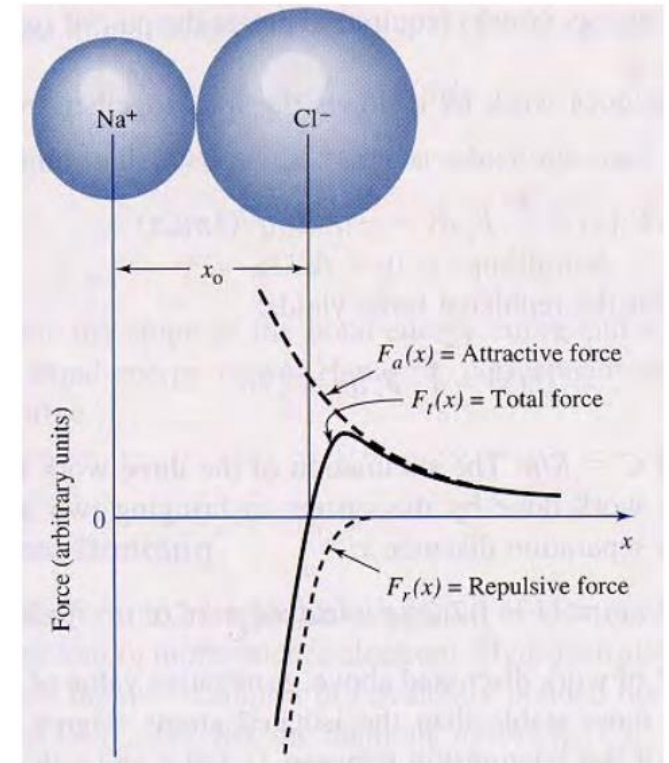
➤ equilibrium spacing:

$$F_A = -\frac{A}{r^p}$$

$$F_R = \frac{B}{r^q}$$

where A, B, p, q are constants; $q > p$

- When $\mathbf{F_A + F_R = 0}$, equilibrium exists.
- The centers of the atoms/ions will remain separated by the **equilibrium spacing** x_0 .

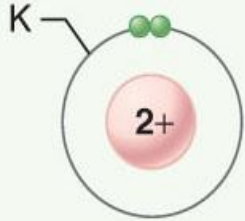
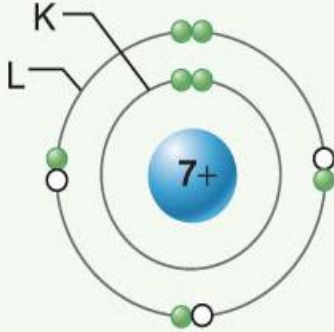


Atomic Bonding in Solids

Bonding Forces & Energies

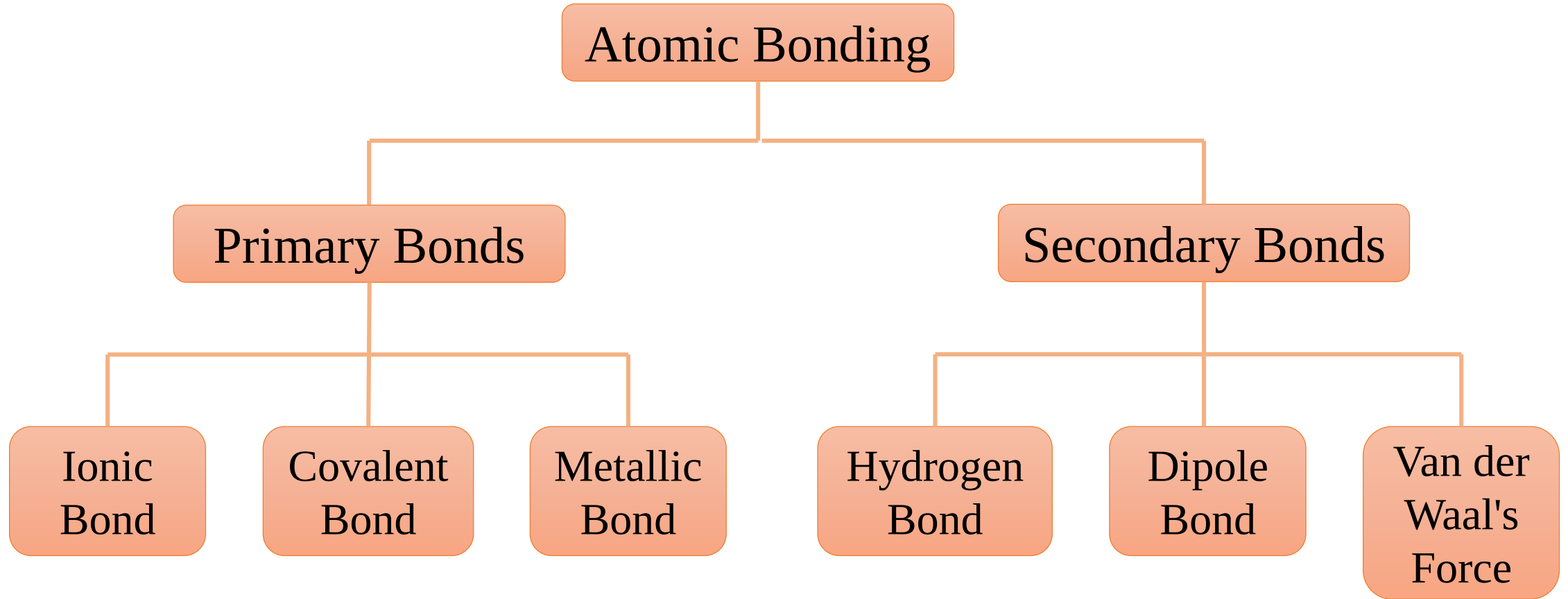
- **Octet rule:** Atoms tend to establish completely-full outer energy levels.
- The *octet rule* is a chemical rule of thumb that reflects the observation that *elements tend to bond* in such a way that each atom has **eight electrons** in its valence shell, giving it the same electronic configuration as a **noble gas**.
- **Atoms** with full energy levels are less reactive than atoms with unfilled energy levels.

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Nonreactive	Reactive
2 protons 2 neutrons 2 electrons	7 protons 7 neutrons 7 electrons
 Helium	 Nitrogen

Atomic Bonding in Solids

Types of Bonding



Atomic Bonding in Solids

Chemical Bonding

- Three different types of *primary* or *chemical bond* are found in solids:
 - 1) ionic,
 - 2) covalent, and
 - 3) metallic.
- *Molecules* are groups of atoms held together in a stable association.
- *Compounds* are molecules containing more than one type of element.
- *Atoms* are held together in molecules or compounds by *chemical bonds*.

Primary Bonding: electrons (e^-) are transferred or shared to form strong bonds;

- **Ionic:**

Example - Na^+Cl

Strong coulomb interaction between a positive atom (lost an electron, Na^+) and a negative atom (an extra electron, Cl^-).

- **Covalent:** electrons *shared* between the atoms. Example - H_2

- **Metallic:** is the electrostatic attractive force b/n conduction electrons and positively charged metal ions. Atoms lose some electrons from valence band, those electrons are shared by *all the material*.

Atomic Bonding in Solids

Chemical Bonding

Ionic Bonding

- Mutual ionization by electron transfer
(remember electronegativity table).
- **Anion** = negatively charged atom
- **Cation** = positively charged atom

Ions are attracted by strong columbic interaction

- Oppositely charged atoms attract
- An ionic bond is non-directional

Example: NaCl

Na has 11 electrons, 1 more than needed for a full outer shell (Neon)

11 Protons Na $1S^2 2S^2 2P^6 3S^1$
11 Protons Na⁺ $1S^2 2S^2 2P^6$

donates e⁻
10 e⁻ left

Cl has 17 electron, 1 less than needed for a full outer shell (Argon)

17 Protons Cl $1S^2 2S^2 2P^6 3S^2 3P^5$
17 Protons Cl⁻ $1S^2 2S^2 2P^6 3S^2 3P^6$

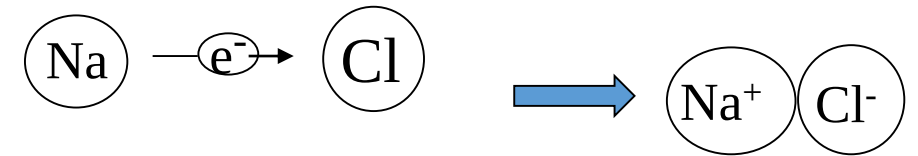
receives e⁻
18 e⁻

Atomic Bonding in Solids

Chemical Bonding

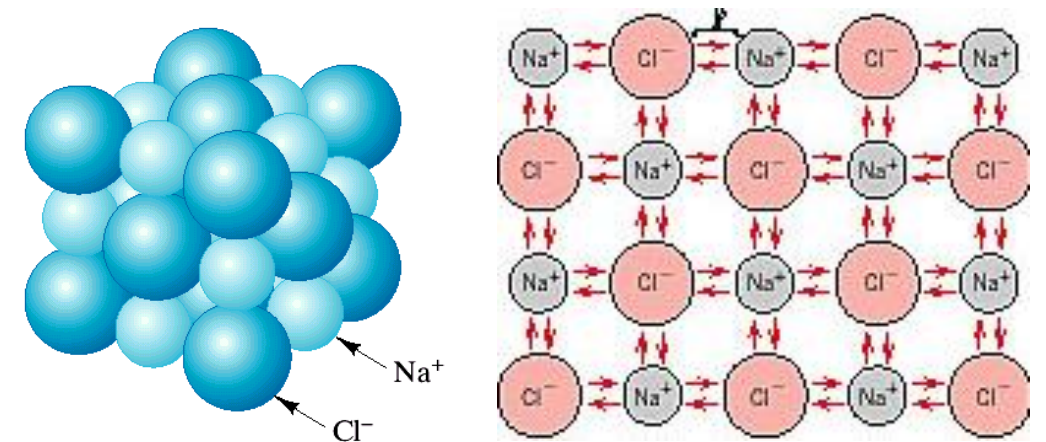
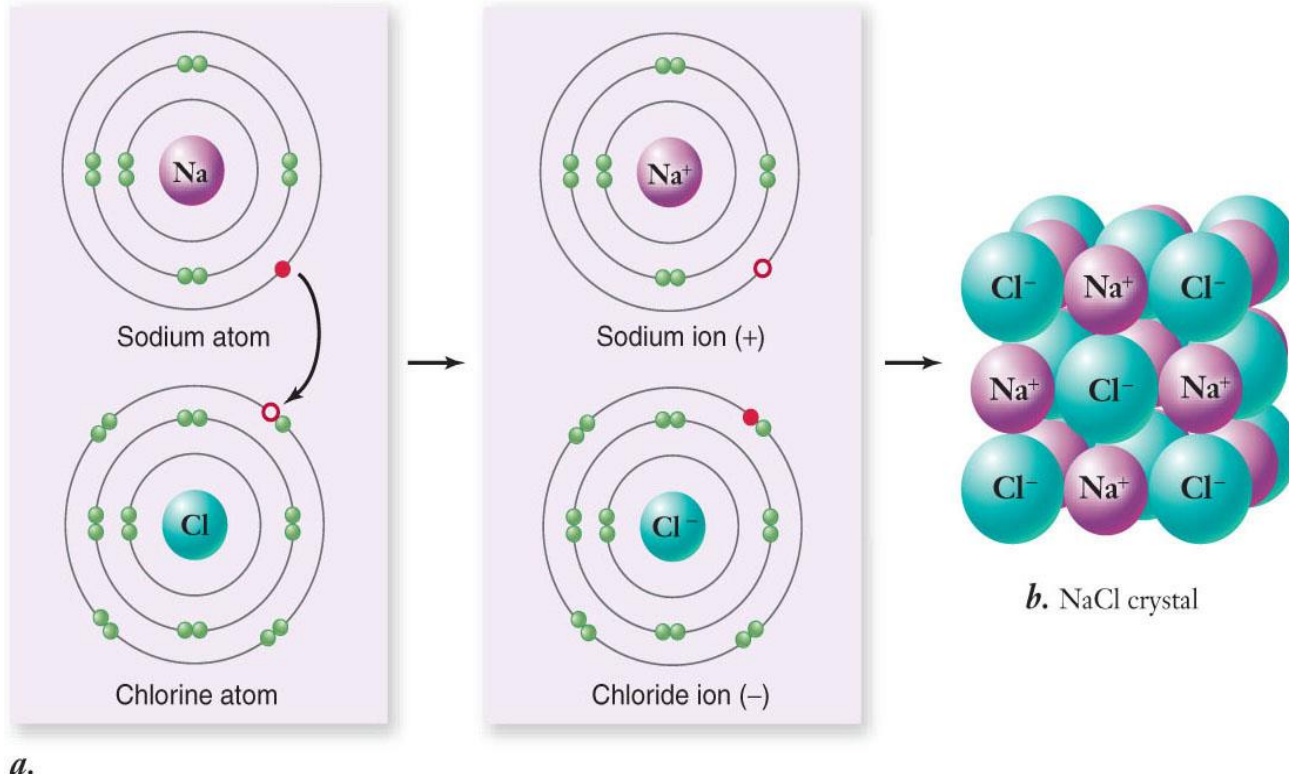
- **Ionic bonds** are formed by the attraction of oppositely charged ions.

- Ionic bonding:



- Electron transfer reduces energy of the system.
- Na shrinks and Cl expands

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Ionic bonds: very strong, nondirectional bonds.

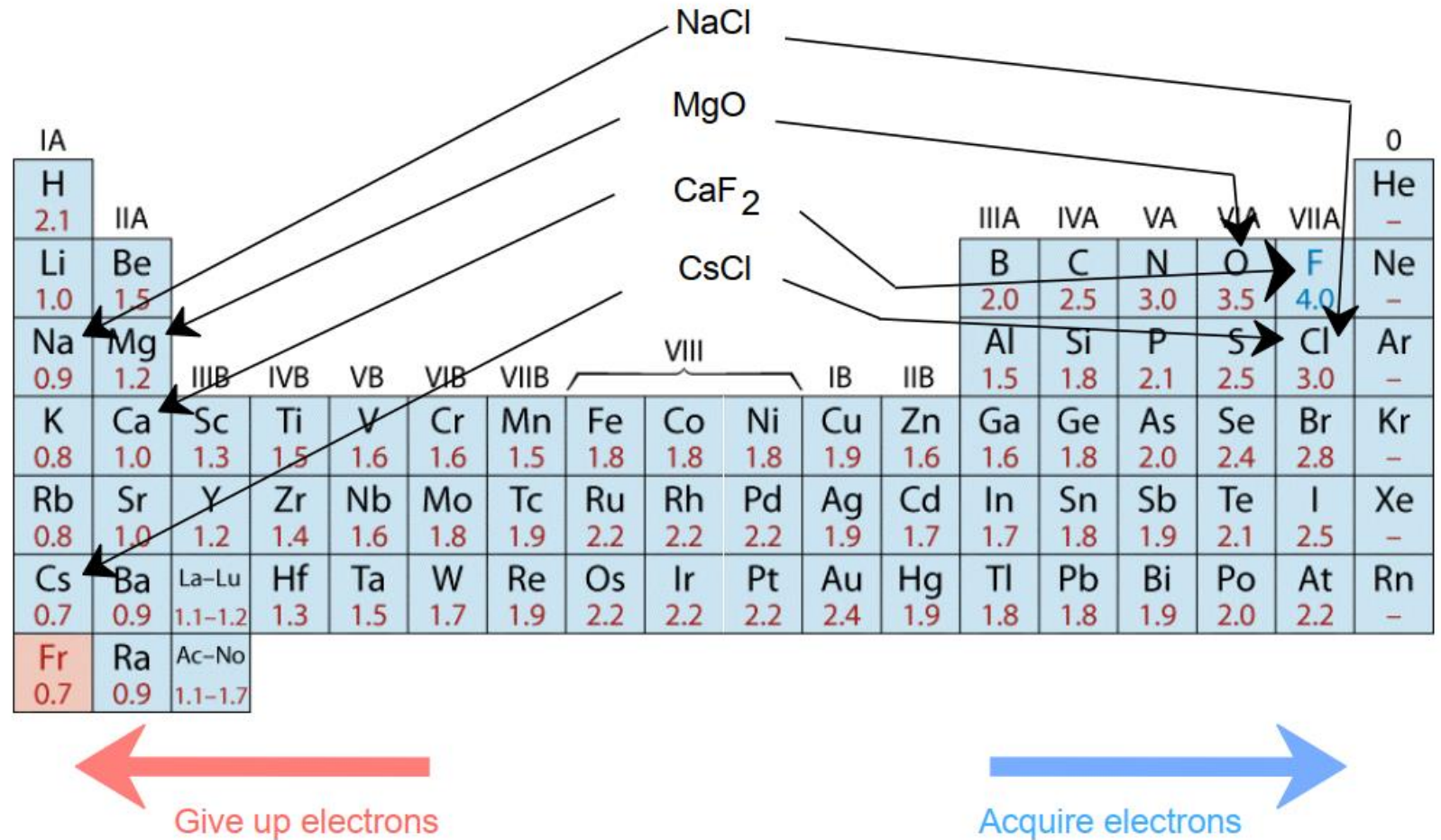
Atomic Bonding in Solids

Chemical Bonding

Ionic Bonding

Examples of ionic bonding

- Predominant bonding in **Ceramics**



Adapted from Fig. 2.7, *Callister 7e*. (Fig. 2.7 is adapted from Linus Pauling, *The Nature of the Chemical Bond*, 3rd edition, Copyright 1939 and 1940, 3rd edition. Copyright 1960 by Cornell University.)

Atomic Bonding in Solids

Chemical Bonding

- The attractive bonding forces are **coulombic**; that is, positive and negative ions, by virtue of their net electrical charge, attract one another.
- For two isolated ions, the attractive energy E_A is a function of the interatomic distance

$$E_A = -\frac{A}{r}$$

An analogous equation for the repulsive energy is

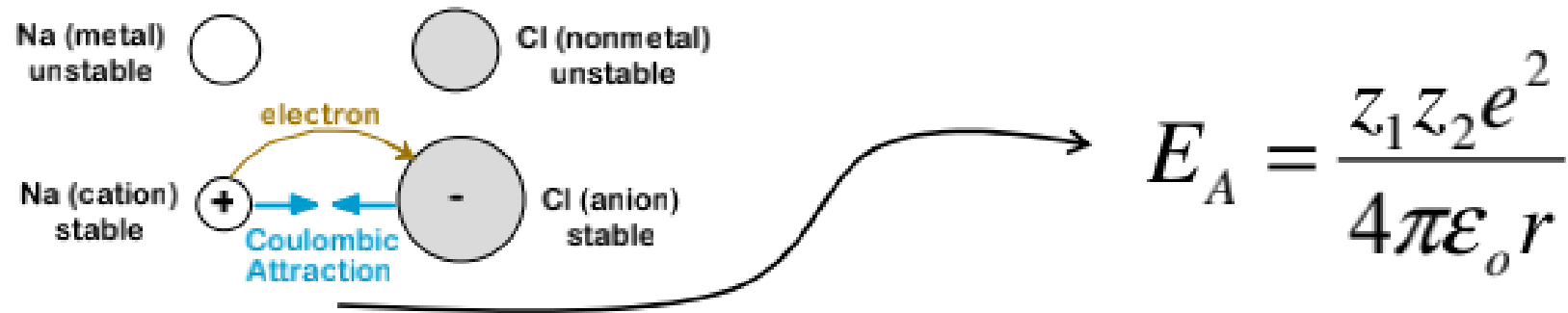
$$E_R = \frac{B}{r^n}$$

The constant **A** is equal to

$$\frac{1}{4\pi\epsilon_0} (Z_1 e)(Z_2 e)$$

- In these expressions, A, B, and n are constants whose values depend on the particular ionic system. The value of n is ~ 8 .
- where ϵ_0 is the permittivity of a vacuum (8.85×10^{-12} F/m) [$C^2 / N.M^2$] Z_1 and Z_2 are the valences of the two ion types, and e is the electronic charge (1.602×10^{-19} C).

Atomic Bonding in Solids



Since $z_1 = +1$ for Na^+ and $z_2 = -1$ for Cl^-

$$E_A = -\frac{e^2}{4\pi\epsilon_0 r} = -\frac{A}{r}$$

Negative energy means attraction only.
Will the atoms collapse on themselves?

No, there is also repulsive energy (e.g. steric repulsion, e-e repulsion)

$$E_R = \frac{B}{r^n}$$

B and n depend on atoms involved.
In many cases $n \sim 8$.

Atomic Bonding in Solids

Ionic bond

Some characteristics of Ionic Bond

Property	Explanation
Melting point and boiling point	The melting and boiling points of ionic compounds are high because a large amount of thermal energy is required to separate the ions which are bound by strong electrical forces.
Electrical conductivity	Solid ionic compounds do not conduct electricity when a potential is applied because there are no mobile charged particles. No free electrons causes the ions to be firmly bound and cannot carry charge by moving.
Hardness	Most ionic compounds are hard; the surfaces of their crystals are not easily scratches. This is because the ions are bound strongly to the lattice and aren't easily displaced.
Brittleness	Most ionic compounds are brittle; a crystal will shatter if we try to distort it. This happens because distortion cause ions of like charges to come close together then sharply repel.

Atomic Bonding in Solids

Chemical Bonding

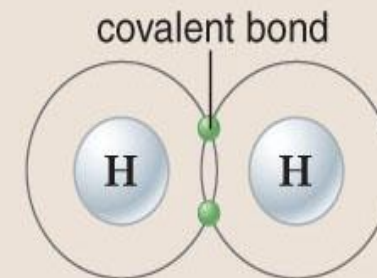
Covalent Bonding

- **Covalent bonds:** form when atoms **share** two or more valence electrons.
- **Covalent bond** strength depends on the number of electron pairs shared by the atoms.

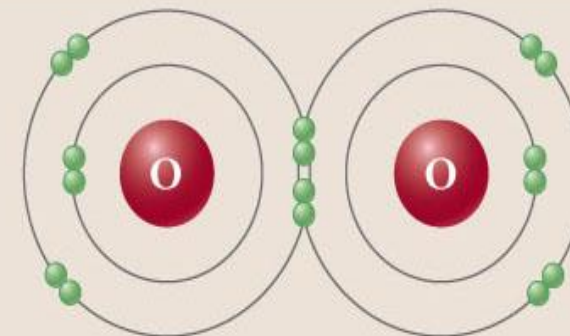
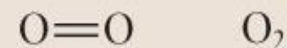
single bond < double bond < triple bond

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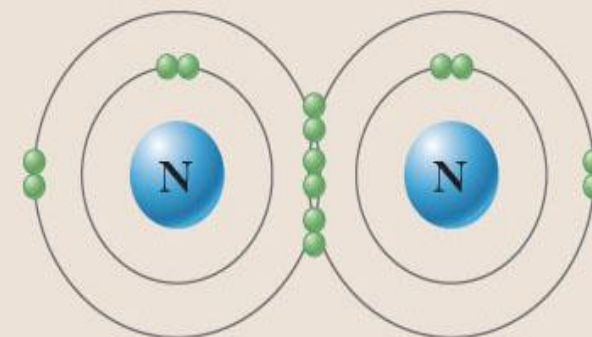
Single covalent bond
hydrogen gas



Double covalent bond
oxygen gas



Triple covalent bond
nitrogen gas



Atomic Bonding in Solids

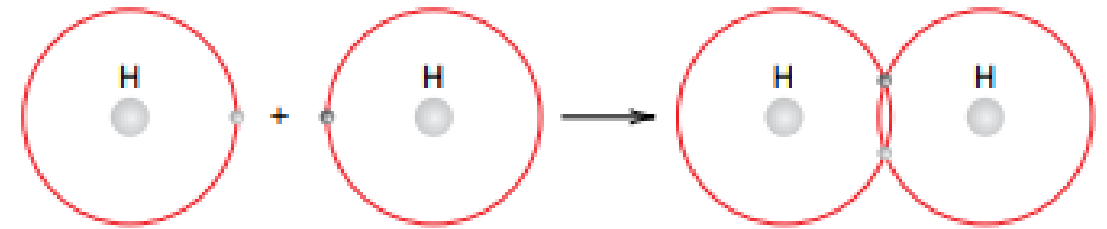
Chemical Bonding

Covalent Bonding

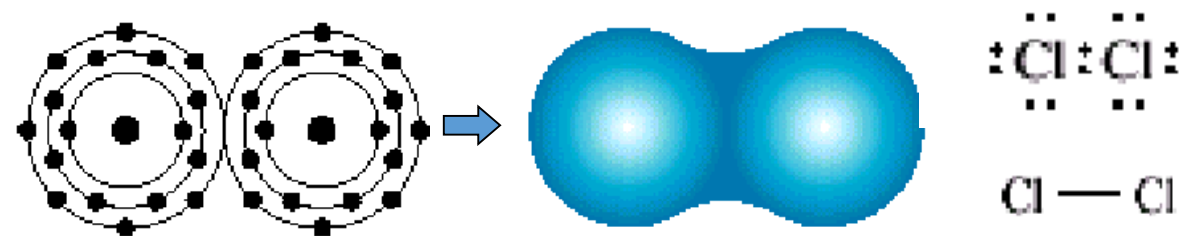
Electronegativity is a chemical property that describes the **tendency** of an atom or a functional group to **attract** electrons toward itself.

- Differences in electronegativity dictate how electrons are distributed in covalent bonds.
- **nonpolar covalent bonds** = equal sharing of electrons.
- **polar covalent bonds** = unequal sharing of electrons.

- **Covalent bonds- highly directional** in direction of greatest orbital overlap.



Example: Cl_2 molecule. $Z_{\text{Cl}} = 17$ ($1s^2 2s^2 2p^6 3s^2 3p^5$)
 $N' = 7$, $8 - N' = 1 \rightarrow$ can form only one covalent bond.



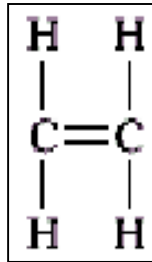
Atomic Bonding in Solids

Chemical Bonding

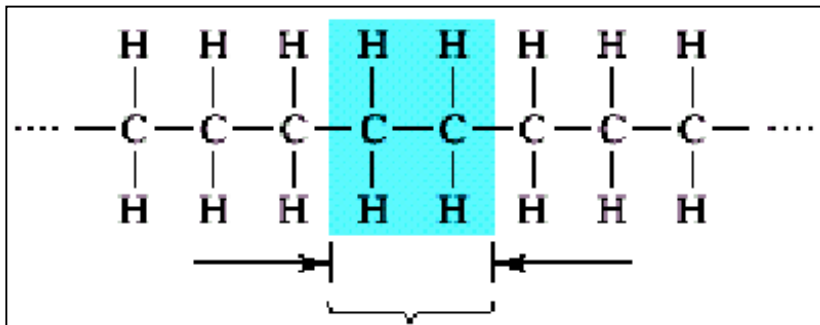
Covalent Bonding

Example: Carbon $Z_c = 6$ ($1s^2 2s^2 2p^2$)
 $N' = 4$, $8 - N' = 4 \rightarrow$ can form up to four covalent bonds.

ethylene
molecule:

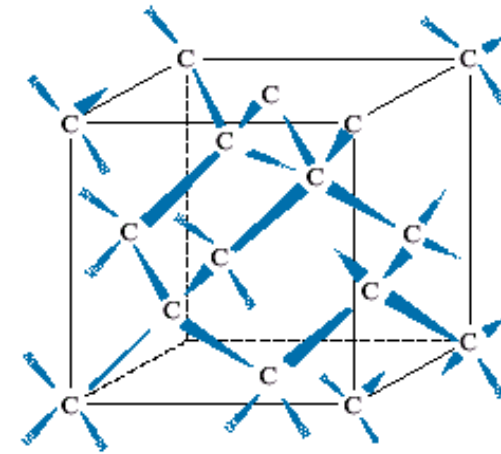


polyethylene molecule:



Ethylene molecule

Diamond: (each C atom has four covalent bonds with four other carbon atoms).



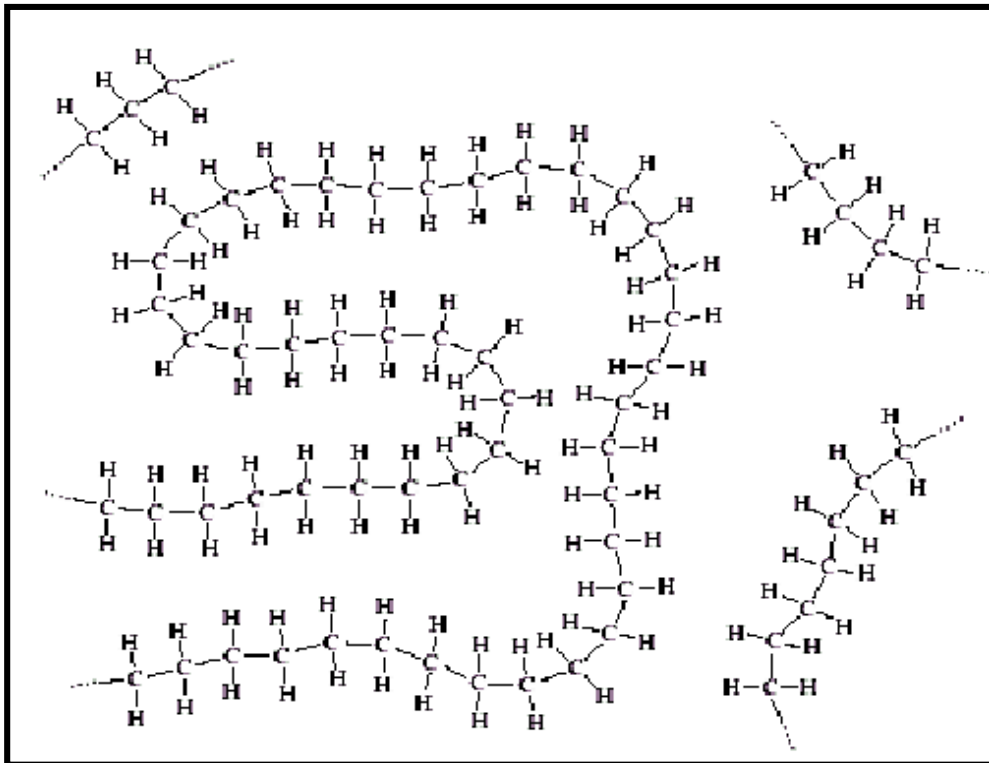
- **Covalent bonds** may be very strong, as in diamond or they may be very weak, as with bismuth.
- **Covalent bonding**, is found in materials whose atoms have small differences in **electronegativity**.

Atomic Bonding in Solids

Chemical Bonding

Covalent Bonding

2-D schematic of the “spaghetti-like” structure of solid polyethylene.



Properties of Covalent Bond

Property	Explanation
Melting point and boiling point	Very high melting points because each atom is bound by strong covalent bonds. Many covalent bonds must be broken if the solid is to be melted and a large amount of thermal energy is required for this.
Electrical conductivity	Poor conductors because electrons are held either on the atoms or within covalent bonds. They cannot move through the lattice.
Hardness	They are hard because the atoms are strongly bound in the lattice, and are not easily displaced.
Brittleness	Covalent network substances are brittle. If sufficient force is applied to a crystal, covalent bonds are broken as the lattice is distorted. Fracture failure occurs rather than deformation of a shape.

Atomic Bonding in Solids

Chemical Bonding

Ionic-Covalent Mixed Bonding

- The closer the atoms are together (i.e., the smaller the difference in electronegativity), the greater the degree of covalency.
- The **percent ionic** character of a bond between elements A and B (A being the most electronegative) may be approximated by the expression
- Most bonds between two different types of atoms are somewhere in between **ionic** and **covalent**. Very few compounds exhibit pure ionic or covalent bonding.

$$\% \text{ ionic character} = \{1 - \exp[-(0.25)(X_A - X_B)^2]\} \times 100$$

where X_A and X_B are the electronegativities for the respective elements.

Metallic - Ionic Mixed Bonding

Typical for intermetallic compounds between elements with significant difference in electronegative

e.g.: NaZn_{13}

Atomic Bonding in Solids

Chemical Bonding

Example:

Calculate the % Ionic Character (IC):

1. MgO, $X_{\text{Mg}} = 1.3$ $X_{\text{O}} = 3.5$

2. NaCl, $X_{\text{Na}} = 0.9$ $X_{\text{Cl}} = 3.0$

Solution

➤ Example: MgO $X_{\text{Mg}} = 1.3$ $X_{\text{O}} = 3.5$

$$\%IC = \left[1 - \exp\left(-\frac{(3.5-1.3)^2}{4}\right) \right] \cdot 100 = [1 - \exp(-1.21)] \cdot 100 = [1 - 0.298] \cdot 100 = 70.2\%$$

➤ Example: NaCl $X_{\text{Na}} = 0.9$ $X_{\text{Cl}} = 3.0$

$$\%IC = \left[1 - \exp\left(-\frac{(3.0-0.9)^2}{4}\right) \right] \cdot 100 = [1 - \exp(-1.10)] \cdot 100 = [1 - 0.332] \cdot 100 = 66.8\%$$

Atomic Bonding in Solids

Chemical Bonding

Metallic Bonding

- Valence electrons are completely **delocalized** to form an electron cloud, in which positive ionic cores are embedded.
- The remaining nonvalence electrons and atomic nuclei form “**ion cores**”, which possess a net positive charge equal in magnitude to the total valence electron charge per atom.
- The metallic bond is **nondirectional**.
- Metallic bonding is found in the periodic table for Group IA and IIA elements.
- Electron delocalization is the origin of **good electrical and thermal conductivities** in metals. (Ionically and covalently bonded materials are typically electrical and thermal insulators, due to the absence of large numbers of free electrons) .

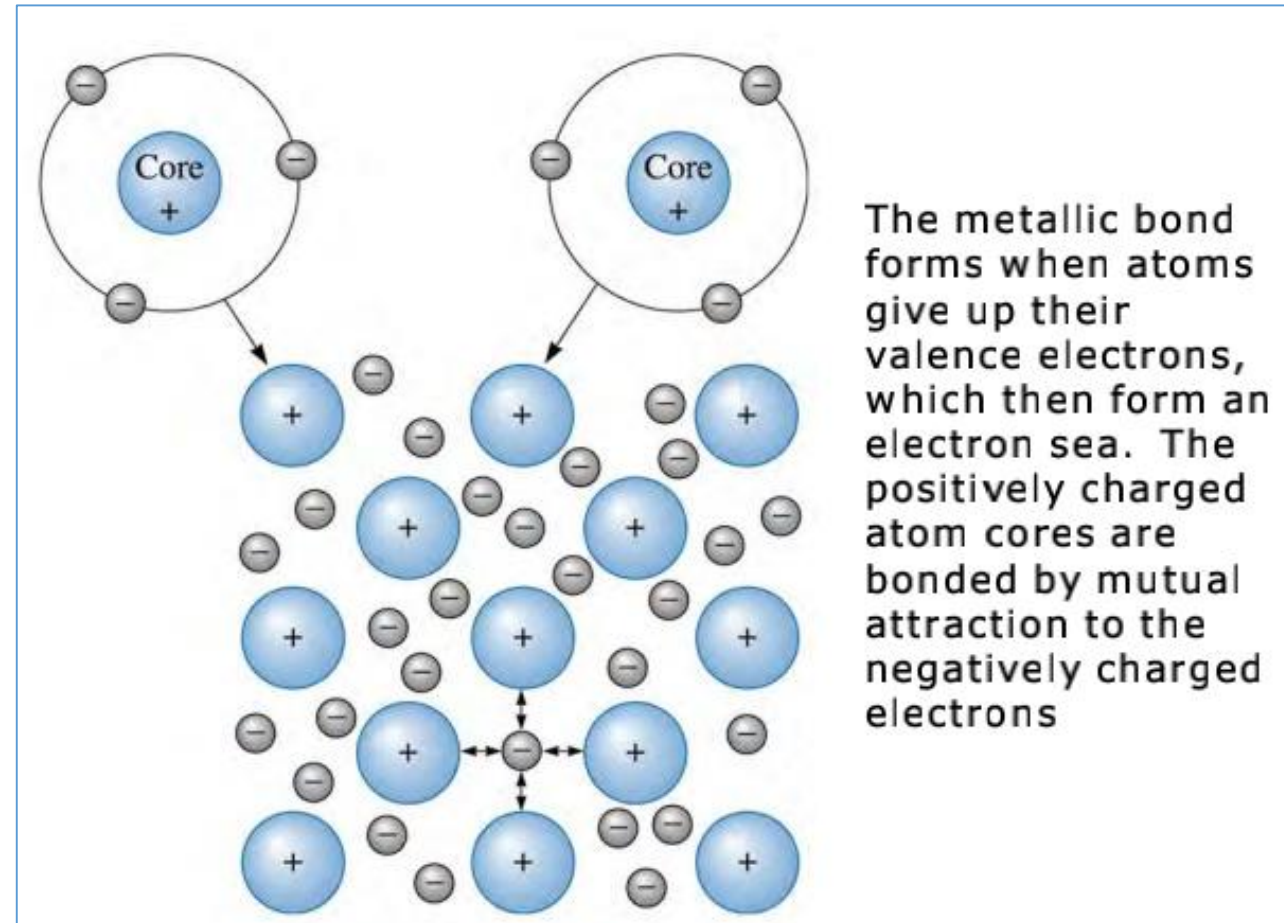
Atomic Bonding in Solids

Chemical Bonding

Metallic Bonding

- Valence electrons are detached from atoms spread in an 'electron sea' that "glues" the “ions” together.
- Metallic bonding, is found in metals and their alloys.
- Metallic bond is non-directional → atoms pack closely.

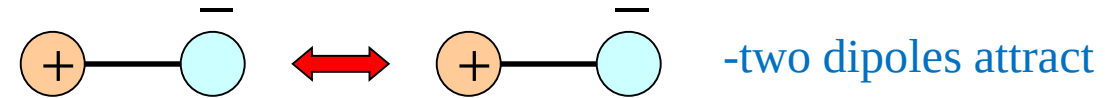
Schematic illustration of metallic bonding.



Atomic Bonding in Solids

Secondary Bonding

- Also known as **physical bonds**
 - **Weak** in comparison to primary or chemical bonds
 - Exist between virtually all atoms and molecules
 - Arise from atomic or molecular dipoles
 - ✓ bonding that results from the **coulombic attraction** between the positive end of one dipole and the negative region of an adjacent one
 - ✓ a **dipole** may be created or induced in an atom or molecule that is normally electrically symmetric
- **Secondary bonding** are not bonds with a *valence electron* being shared or donated.
 - They occur due to forces produced by atomic or molecular **dipoles**.

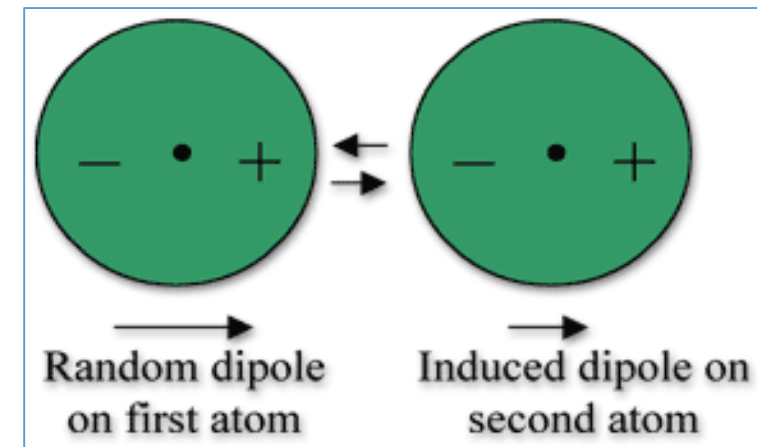
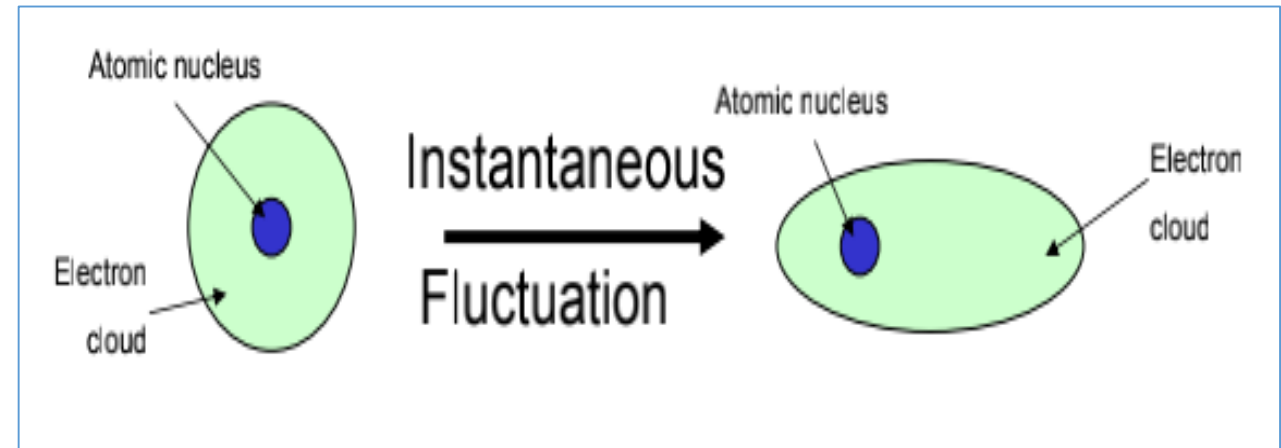


Atomic Bonding in Solids

Secondary Bonding

Fluctuating Induced Dipole Bonds

- A dipole (whether induced or instantaneous) produces a displacement of the electron distribution of an adjacent molecule or atom and continues as a chain effect
- Liquefaction and solidification of inert gases
- **Weakest** Bonds
- Extremely low boiling and melting point

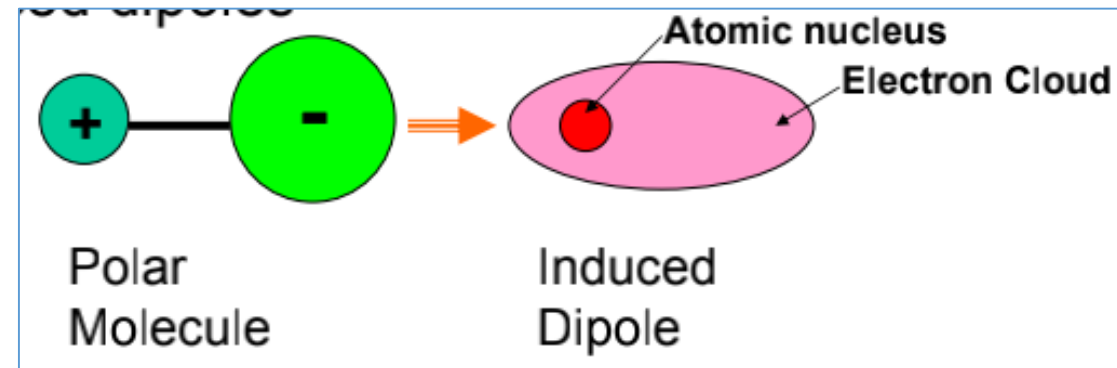


Atomic Bonding in Solids

Secondary Bonding

Polar Molecule-Induced Dipole Bonds

- Permanent dipole moments exist by virtue of an asymmetrical arrangement of positively and negatively charged regions
- Polar molecules can induce dipoles in adjacent nonpolar molecules
- Magnitude of bond **greater** than for **fluctuating** induced dipoles

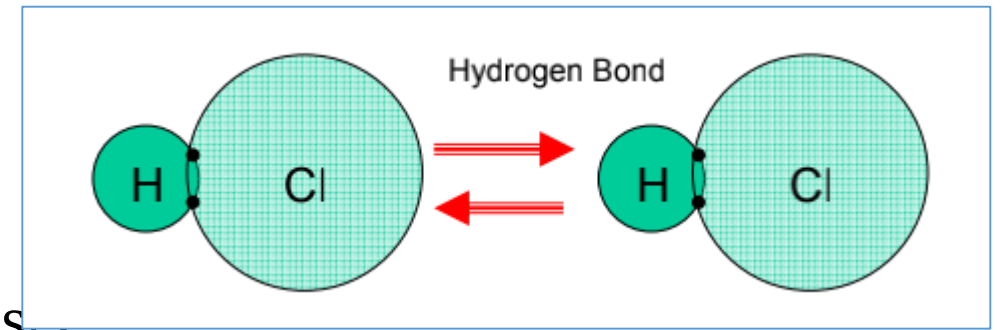


Atomic Bonding in Solids

Secondary Bonding

Permanent Dipole Bonds

- **Stronger** than any secondary bonding with induced dipoles
- A special case of this is hydrogen bonding: exists between molecules that have hydrogen as one of the constituents



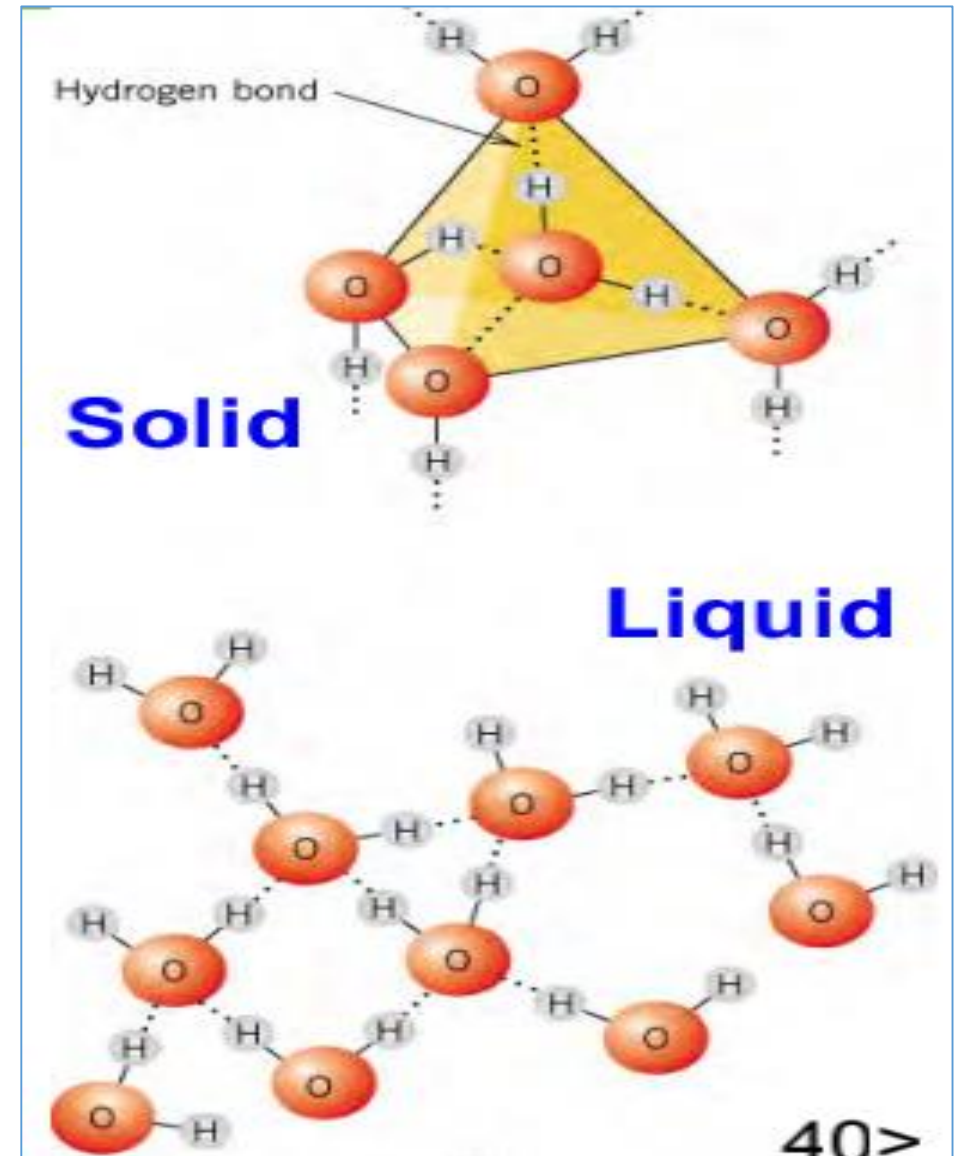
Atomic Bonding in Solids

Secondary Bonding

Permanent Dipole Bonds

Hydrogen bonds in water

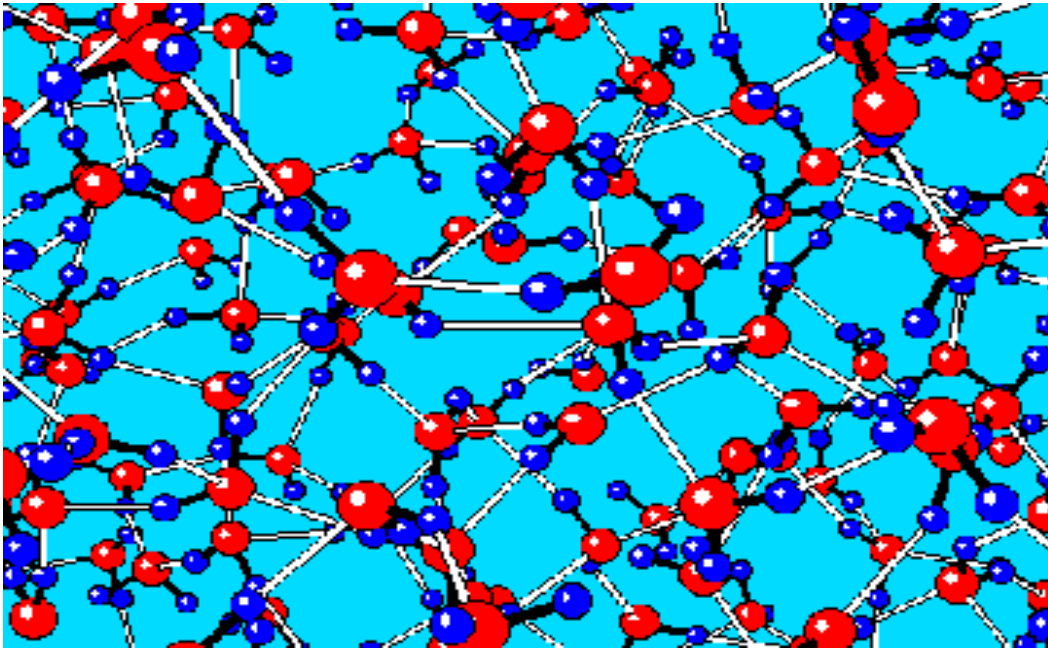
- Many molecules do not have a symmetric distribution/arrangement of positive and negative charges (e.g. H_2O , HCl , NH_3)



Atomic Bonding in Solids

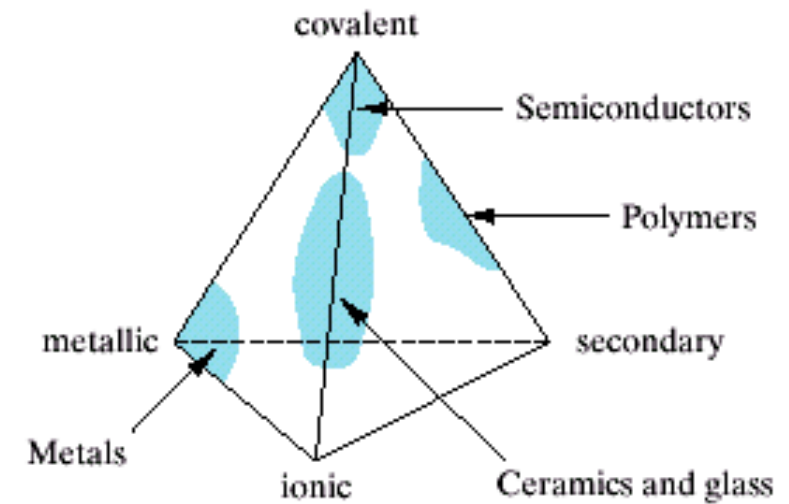
Secondary Bonding

- Hydrogen bonding in liquid water from a molecular-level simulation.



Molecules: Primary bonds inside, secondary bonds among each other.

Summary



- Examples of bonding in Materials:

Metals: Metallic

Ceramics: Ionic / Covalent

Polymers: Covalent and Secondary

Semiconductors: Covalent or Covalent/Ionic

Summary

Table 2.3 Bonding Energies and Melting Temperatures for Various Substances

Bonding Type	Substance	Bonding Energy		Melting Temperature (°C)
		<i>kJ/mol</i> (<i>kcal/mol</i>)	<i>eV/Atom,</i> <i>Ion, Molecule</i>	
Ionic	NaCl	640 (153)	3.3	801
	MgO	1000 (239)	5.2	2800
Covalent	Si	450 (108)	4.7	1410
	C (diamond)	713 (170)	7.4	>3550
Metallic	Hg	68 (16)	0.7	−39
	Al	324 (77)	3.4	660
	Fe	406 (97)	4.2	1538
	W	849 (203)	8.8	3410
van der Waals	Ar	7.7 (1.8)	0.08	−189
	Cl ₂	31 (7.4)	0.32	−101
Hydrogen	NH ₃	35 (8.4)	0.36	−78
	H ₂ O	51 (12.2)	0.52	0

Atomic Bonding summary

Type	Bond Energy	Comments
Ionic	Large!	Nondirectional (ceramics)
Covalent	Variable large-Diamond small-Bismuth	Directional (semiconductors , ceramics polymer chains)
Metallic	Variable large-Tungsten small-Mercury	Nondirectional (metals)
Secondary	smallest	Directional inter-chain (polymer) inter-molecular

Exercises

- 1) Zinc has five naturally occurring isotopes: 48.63% of ^{64}Zn , with an atomic weight of 63.929 amu; 27.90% of ^{66}Zn , with an atomic weight of 65.926 amu; 4.10% of ^{67}Zn , with an atomic weight of 66.927 amu; 18.75% of ^{68}Zn , with an atomic weight of 67.925 amu; and 0.62% of ^{70}Zn , with an atomic weight of 69.925 amu. Calculate the average atomic weight of Zn.
- 2) Indium has two naturally occurring isotopes: ^{113}In , with an atomic weight of 112.904 amu; and ^{115}In , with an atomic weight of 114.904 amu. If the average atomic weight for In is 114.818 amu, calculate the fraction-of-occurrences of these two isotopes.
- 3) The force of attraction between a divalent cation and a divalent anion is 1.67×10^{-8} N. If the ionic radius of the cation is 0.080 nm, what is the anion radius?
- 4)
 - (a) Calculate the coulombic attractive force between a pair of Na^+ and Cl^- ions that just touch each other. Assume the ionic radius of Na^+ ion to be 0.095nm and that of the Cl^- ion to be 0.181nm.
 - (b) Calculate the net potential energy of a simple Na^+Cl^- ion pair by using the following expression
$$E_{\text{net}} = E_A + \frac{b}{r^n}, \text{ where } n=9 \text{ for NaCl}$$
- 5) Order the following semiconductors from most covalent to most ionic.
 - 1) ZnS, GaP, CuCl
 - 2) ZnS, ZnSe, ZnO

Summary Questions

- *Why periodic table is important in studying engineering materials?*
- *Show the electron configuration for Li, Ne, Mg, Ar, K, Ca (ion), Cu, Fe, I and specify the group, period and block naming.*
- *What is the difference between atomic mass and atomic weight?*
- *Briefly explain the main differences among ionic, covalent, and metallic bonding.*
- *State the Bohr model and Pauli exclusion principles.*
- *What type(s) of bonding would be expected for brass?*
- *What type(s) of bonding would be expected for rubber?*
- *What are the four quantum numbers and what they specify relative to electrons and electron states?*
- *Calculate the force of attraction between a Ca ion and an O ion whose centers are separated by a distance of 1.25 nm.*

End of the Chapter!

Thank you!